

CX10K Hands on Lab User Guide

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Doc version

0.91	AMD / Pensando Team	New Tasks added	5/16/2025
1.0	AMD / Pensando Team	Image updates	6/6/2025

Key Abbreviations

These will help you reference any abbreviations you encounter while reading through the lab Guide

- DPU = Data Processing Unit
- vDS = VMWare Distributed Switch
- DSM = Distributed Services Module
- DSS = Distributed Services Switch (the CX 10000)
- ELK = Elasticsearch Logstash Kibana

PSM = Policy and Services Manager

VM = Virtual Machine

WS = Workstation

Learning Objectives

Upon completion of this Hand's on Lab you will be able to:

1. Understand the value and benefit of a DPU enabled smart switch
2. Interpret VMware networking, PSM policy and ELK visibility
3. Familiarize yourself with the layout of PSM and ELK interfaces
4. Understand how micro- and macro-segmentation can be achieved on east/west workloads on the CX 10000 which can help you achieve your Zero Trust architecture goals in your datacenter architecture
5. Have a working knowledge of basic base policy features of PSM

Getting Started

This guide provides a high-level overview of the capabilities of the CX 10000 Distributed Services Switch (DSS). The purpose of this guide is to provide guidance and inspiration for experimenting within your own controlled, self-contained lab environment either through the PSM or via the API. Since each hosted environment of the DPU test-drive will vary, the course leader will explain any local variations or changes. This course aims to familiarize you with the features that can be configured and managed via PSM. Please **explore**, break, learn, and enhance your knowledge of the products during this time.

Scenario

Throughout this lab activity, you will learn how the HPE Aruba CX1000 with AMD Pensando DPU solution can enable a next gen data center architecture.

By inserting stateful service capabilities in a data center fabric, security and visibility are now integrated into your environment and placed closer to where applications and workloads live without any impact on existing network architecture or software configuration changes.

In this lab you will learn how to review and understand VMware networking, set up macro & micro-segmentation, firewalling services policy through the Pensando Policy and Services Manager (PSM) and gain visibility to your data in the Elastic - ELK stack.

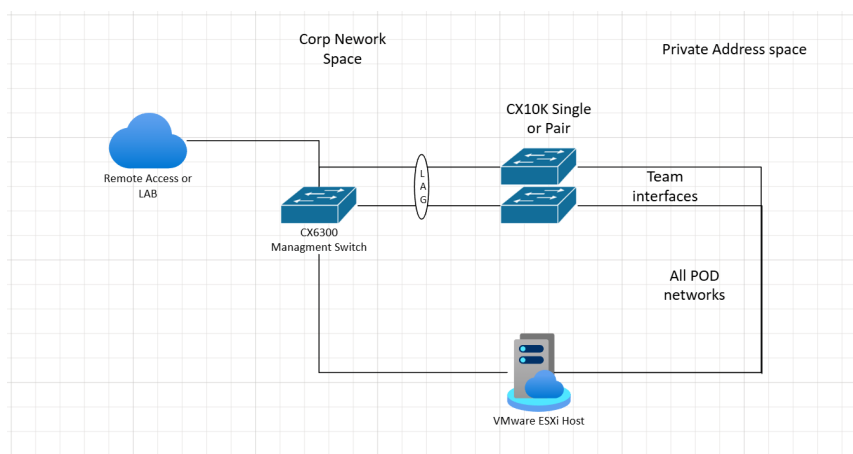
The CX10k includes a pair of Pensando/AMD DPU's. While these DPUs support multiple programmable functions like stateful firewalling, encryption, NAT, and real-time telemetry, this lab focuses on firewalling, security and policy enforcement.

PSM = Pensando Services manager is a programmable, secure and highly available centralized system for managing infrastructure policy. It is designed to establish and manage policies for every CX 10000 Distributed Services Switch (DSS).

ELK = Is a suite of open-source tools for log management, analysis, and visualization comprised of Elasticsearch, Kibana, Beats, and Logstash (also known as the ELK Stack). It allows you to take data from the CX 10000 switch in any format, then search, analyze, and visualize that data.

Topology

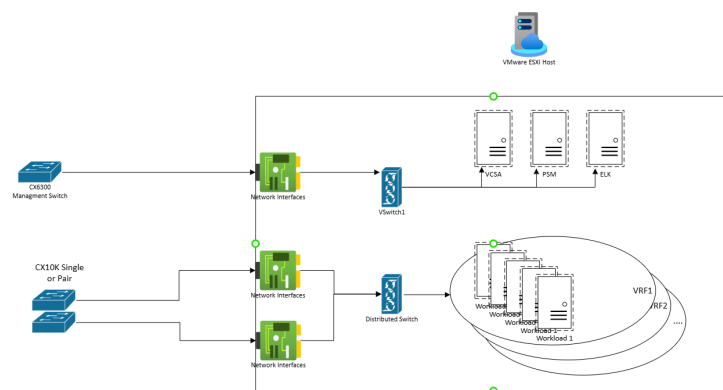
Understanding the Lab Environment



The above diagram shows an overview of the lab topology; the actual hardware might change, but it is shown as an example.

The lab is designed so that all pod access is controlled, and you have a safe environment to work and learn. Access to the environment can be provided by either remote access or classroom access (when onsite). Everything you access will be hosted on the VMware ESXi host. Your access to the services on the ESXi host is via the management switch.

The VMware ESXi has 3 network interfaces configured. One – north to the management switch where your access is granted, and then Two – southbound to the CX10k that are used for the services testing.



The applications you have access to as part of the LAB are the vCenter server (VCSA), PSM, and ELK. Via the vCenter server, you can access your pod and configure your workloads as required.

The pod has been set up with a default configuration; you will be required to access the workloads and configure them as you require to carry out the tests. It is expected that you will use applications like PING/SSH/CURL/IPERF to test the features.

The lab will take you through things like the moving of the workloads to a PVLAN Isolation network to perform “micro segmentation”, and/or moving of the workloads to a different VLAN to perform “macro segmentation”.

The lab guide has been written as an example and to explain what is available, but it is really to grant you free access to test the scenarios that are important to you. Please feel free to expand the lab test cases to meet your own requirements.

Username

The username you have been assigned is based on your pod ID. This allows for segmentation of your environment from other students. However, as this is running on

shared infrastructure for the vSphere, PSM and ELK environments, we ask that everyone stays within their own pod.

To access any items specific to your pod simply replace the <POD> in the examples with your given assigned pod ID.

Example:

POD	PSM user	VCenter user	ELK User
1	vrf1_user	vrf1_user@vsphere.local	vrf1_user
2	vrf2_user	vrf2_user@vsphere.local	vrf2_user
....			

PSM Username

There are two accounts below that allow you to login to PSM. The read-only account grants access to the entirety of PSM, while the second account has restricted write permissions, allowing you to modify your network and policies within your pod.

The read-only account is optional for exploration of all that PSM has to offer, which can be useful to better understand the layout and options available along with how the Role Based Access works within PSM.

Username	Password	Role
readonly	Pensando0\$	Read-only
vrf<pod>_user	Pensando0\$	vrf(pod)

PSM Role Read-only.

The read-only role has full read permissions to all objects inside of PSM; it does not have any write, update, or delete permissions.

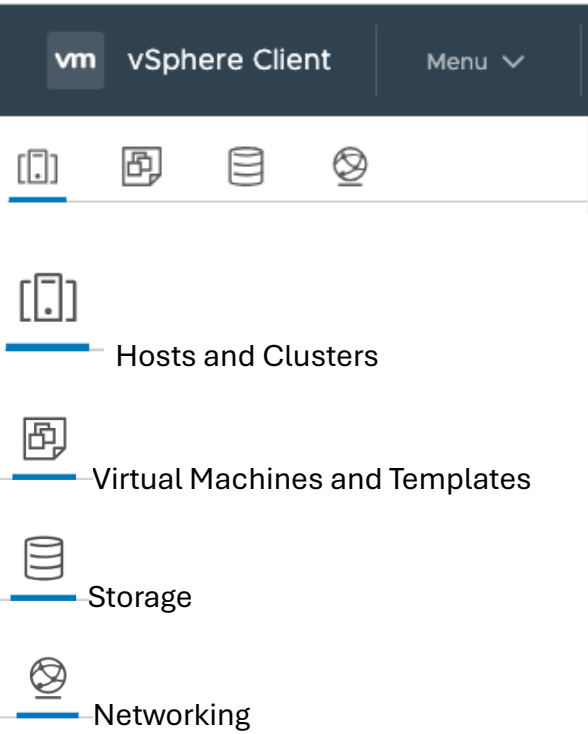
PSM Role vrf (pod)

The (pod) role has full read access to most of the object tree. However, for VRF, Networks, and Security policies, it only has read-write permissions for the assigned objects.

You will primarily use the vrf<pod>_user account throughout the lab. However, in some cases you may want to log in with a full read-only account. When doing so, it is best to use an incognito window.

VMware

For those unfamiliar with the vSphere client, the symbols below represent the main areas you will be working in. From left to right they are:



You will also receive credentials to access a shared vCenter server environment. This account will have permissions scoped to the Port groups, Resource Pools, and virtual machine workloads assigned to you.

Username	Password	Role
vrf<pod>_user@vsphere.local	Pensando0\$	TestDrive
readonly@vsphere.local	Pensando0\$	Read only

Role TestDrive.

The “TestDrive” role includes the following assigned permissions:

Distributed switch

- Move

ESX Agent Manager

- View

Host

- Configuration
 - Change settings

vCenter Server Profiles

- vCenter Server Profiles Read Privileges

privilege.IntercomNamespace.label

- privilege.IntercomNamespace.Read.Label

vSphere Tagging

- Assign or Unassign vSphere Tag
- Assign or Unassign vSphere Tag on Object

Namespaces

- Modify namespace configuration

Network

- Assign network
- Configure
- Move network

Virtual machine

- Change Configuration
 - Add new disk
 - Advanced configuration
 - Change Settings
 - Display connection settings
 - Modify device settings
- Edit Inventory
 - Move
- Interaction
 - Console interaction
 - Pause or Unpause
 - Power off
 - Power on
 - Reset
 - Suspend

VMware Role read-only

The read-only role has full read permissions to all objects inside vSphere; it does not have any write, update or delete permissions.

Workstations (VM’s)

VM Accounts

Each pod is provisioned with Five workstations. These VM’s are initially assigned IP addresses from the primary network range and vlan but may be moved or migrated across different networks to test and validate various policy changes and configurations.

User	Password
tc	VMware1!
root	VMware1!

Commands to edit a VM's IP

To change the IP address of the workstation, the following command options are available:

Option 1

```
sudo ifconfig eth0 <IP> netmask <mask> up
sudo route add default gw <gateway>

example.

sudo ifconfig eth0 192.168.11.11 netmask 255.255.255.0 up
sudo route add default gw 192.168.11.1
```

Option 2

```
sudo /opt/set-ipv4-address.sh <IP> <mask> <broadcast> <Gateway> <DNS> <domain>

example.

sudo /opt/set-ipv4-address.sh 192.168.11.11 255.255.255.0 192.168.11.255
192.168.11.1 192.168.101.1 testdrive
```

VM Networks and subnets

Each pod has been allocated three VLANs, and three subnet ranges specific to the pod.

The model for the networks and subnets is based on the pod number.

VRF – vrf<pod>

Network	Subnet	Mask	Gateway	DNS	Domain
VLAN<pod>1	192.168.<pod>1.0	/24	192.168.<pod>1.1	8.8.8.8	Testdrive
VLAN<pod>2	192.168.<pod>2.0	/24	192.168.<pod>2.1	8.8.8.8	Testdrive
VLAN<pod>3	192.168.<pod>3.0	/24	192.168.<pod>3.1	8.8.8.8	Testdrive

For example, Pod 1.

VRF – vrf1

Network	Subnet	Mask	Gateway	DNS	Domain
VLAN11	192.168.11.0	255.255.255.0	192.168.11.1	8.8.8.8	Testdrive
VLAN12	192.168.12.0	255.255.255.0	192.168.12.1	8.8.8.8	Testdrive

VLAN13	192.168.13.0	255.255.255.0	192.168.13.1	8.8.8.8	Testdrive
--------	--------------	---------------	--------------	---------	-----------

Workstation Addresses

Each pod has three workstations deployed on the first network and the following Port-Group

VRF – vrf<pod>			
Workstation	IP Address	Port Group	Example
Workstation<pod>1_11	192.168.<pod>1.11	<pod>1_10<pod>1_PRO	Workstation11_11 / 192.168.11.11 / 11_1011_PRO
Workstation<pod>1_12	192.168.<pod>1.12	<pod>1_10<pod>1_PRO	Workstation11_12 / 192.168.11.12 / 11_1011_PRO
Workstation<pod>1_13	192.168.<pod>1.13	<pod>1_10<pod>1_PRO	Workstation11_13 / 192.168.11.13 / 11_1011_PRO

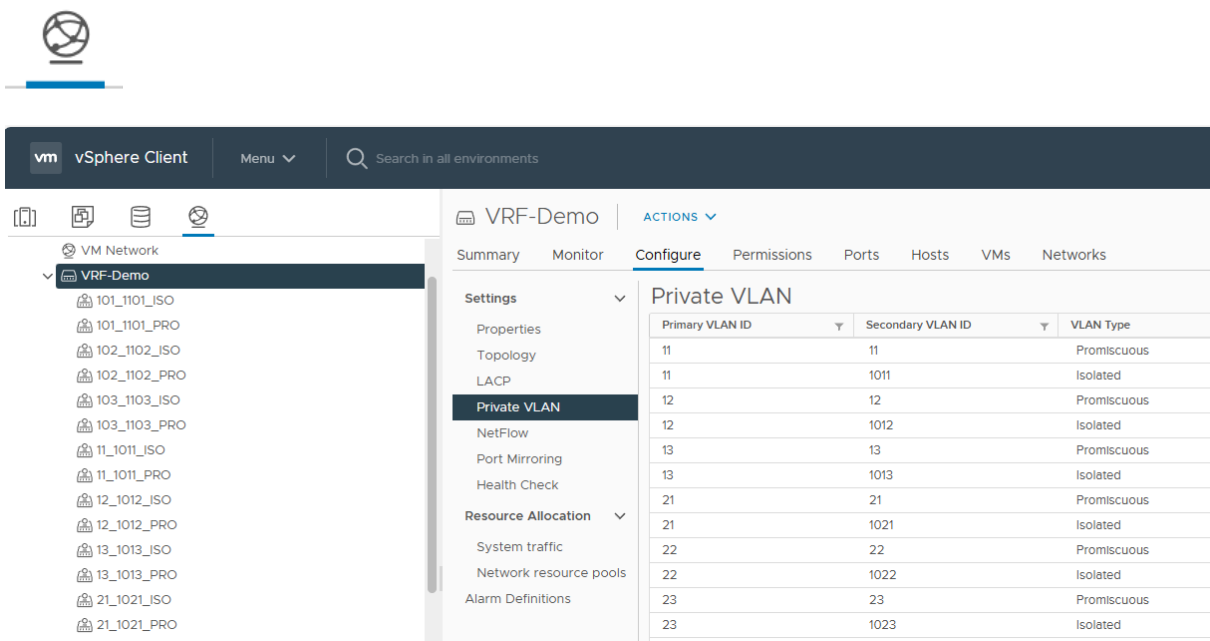
VMWare Networking Exploration

The VMware Distributed Switch and Port groups have been pre-configured; each individual pod space has been allocated a total of 6 Port groups with 2 port groups per vlan based on a standard PVLAN configuration. In the micro-segmentation scenario, we use PVLAN isolation to get the traffic from the Distributed switch port group out and up to the CX 10000, allowing traffic to be visualized and managed.

The “Distributed switch” created is called “VRF-DEMO”, and the configuration can and should be reviewed with the read-only account to gain a better understanding of the environment. (use the readonly@vsphere.local account to review)

vSphere Distributed Switch

In the **Networking** tab under your provisioned DVS in the configure section of the distributed switch you can find the Private VLAN section. Each primary Network VLAN ID will require a secondary VLAN configured under it and selected for Isolated mode.



In this setup, Isolated mode is used to direct traffic to the CX10k’s SVI interfaces for management. While community mode is an available option for PVLAN configuration, it is not used in this scenario.

vSphere Port-Groups

The configuration of the Isolated and Primary networks must align with the network settings on the CX10k’s.

Primary VLAN	Secondary VLAN	Primary Port Group	Isolation Port Group
<pod>1	10<pod>1	<pod>1_10<pod>1_PRO	<pod>1_10<pod>1_ISO
<pod>2	10<pod>2	<pod>2_10<pod>2_PRO	<pod>2_10<pod>2_ISO
<pod>3	10<pod>3	<pod>3_10<pod>3_PRO	<pod>3_10<pod>3_ISO

*PRO designates a promiscuous vlan type

*ISO designates an isolated vlan type

Example

Primary VLAN	Secondary VLAN	Primary Port Group	Isolation Port Group
11	1011	11_1011_PRO	11_1011_ISO
12	1012	12_1012_PRO	12_1012_ISO

13	1013	13_1013_PRO	13_1013_ISO
----	------	-------------	-------------

PVLAN Modes

Promiscuous	A port that is a member of the primary VLAN. These ports can send packets to all ports of the primary VLAN and ports of associated isolated and community VLANs. Promiscuous ports are used to communicate outside the PVLAN domain.
Isolated	A secured VLAN where hosts cannot communicate with each other through L2 switching.
Community	VLAN where hosts can communicate with each other through L2 switching. There can be multiple community VLANs associated with a primary VLAN. CX 10000 does not support this PVLAN mode.

PSM

The majority of your tasks will be performed through PSM, the management platform responsible for the security components of the CX 10000.

The PSM can be deployed as an OVA image on VMware, VHD on Hyper-V, or qcow image on any KVM based platform, as previously mentioned, it has already been deployed for you in this lab environment, and the two DSSes (CX 10000s) have already been admitted to PSM. You can see this in PSM explore the PSM UI and find where the switches are healthy and admitted. Hint: DSS.

 For more information, options, and a quick walkthrough on PSM, refer to Appendix A or visit the user-guide at <https://networkingsupport.hpe.com/>.

PSM VRF and Network

To match the VLANs configured on the CX 10000 switch and the VMware environment, every pod is allocated 5 networks in PSM within the dedicated pod VRF you have been assigned.

The networks within PSM are linked to the CX 10000 based on the VLAN ID, VRF ID and must match the primary (Promiscuous VLAN) in the PVLAN configuration on the VMware environment.

VRF – vrf<pod>

Network	VLAN ID	VRF	Ingress Policy	Egress Policy
VLAN<pod>1	<pod>1	vrf<pod>		
VLAN<pod>2	<pod>2	vrf<pod>		
VLAN<pod>3	<pod>3	vrf<pod>		

Example from Pod 1

VRF – vrf1

Network	VLAN ID	VRF	Ingress Policy	Egress Policy
VLAN 11	11	Vrf1		Vrf1_vlan11_First
VLAN 12	12	Vrf1		Vrf1_vlan12_First
VLAN 13	13	Vrf1		Vrf1_vlan13_First

PSM Security Policies

There have been 10 pre-defined security policies created for each pod to begin your work with; they can be assigned to either Egress or Ingress on the VRF or Network.

The names allocated to the policies as part of the lab are available to use in any way you see fit. It is recommended that you use the first policy (labeled _First) as the active policy and your newly created policy as the policy you will migrate to your environment.

Policies	
vrf<pod>_vlan<pod>1_First	vrf<pod>_vlan<pod>1_new
vrf<pod>_vlan<pod>2_First	vrf<pod>_vlan<pod>2_new
vrf<pod>_vlan<pod>3_First	vrf<pod>_vlan<pod>3_new

Example from Pod 1

Policies		Example Pod 1	
Vrf1_vlan11_First	Vrf1_vlan11_new	vrf1_vlan11_First	vrf1_vlan11_new
Vrf1_vlan12_First	Vrf1_vlan12_new	vrf1_vlan12_First	vrf1_vlan12_new

Vrf1_vlan13_First	Vrf1_vlan13_new	vrf1_vlan13_First	vrf1_vlan13_new
-------------------	-----------------	-------------------	-----------------

 Additional information regarding Policy enforcement from a PSM/DPU perspective regarding ingress and egress definitions can be found here Pg.94
https://arubanetworking.hpe.com/techdocs/Pensando/AMD_Pensando_PSM_for_DSS_Guide-1.100.2-T.pdf

Activities

The URL for the Vcenter server will be provided by the trainer, as it is unique to every lab.

Activity: Exploring VMware networking

Review The VMware configuration using the “read-only” account, to get a better understanding of the layout.

Username	Password	Role
readonly@vsphere.local	Pensando0\$	Read only



1. Go to the **network tab** and select the “VRF-Demo” Distributed switch.
 - a. Look at the configure – Private VLAN section, you will see the VLAN mappings.

VRF-Demo

ACTIONS

SummaryMonitorConfigurePermissionsPortsHostsVMsNetworks

Settings

Properties

Topology

LACP

Private VLAN

NetFlow

Port Mirroring

Health Check

Resource Allocation

System traffic

Network resource pools

Alarm Definitions

Private VLAN

Primary VLAN ID	Secondary VLAN ID	VLAN Type
11	11	Promiscuous
11	1011	Isolated
12	12	Promiscuous
12	1012	Isolated
13	13	Promiscuous
13	1013	Isolated
21	21	Promiscuous
21	1021	Isolated
22	22	Promiscuous
22	1022	Isolated
23	23	Promiscuous
23	1023	Isolated
24	24	Promiscuous

b. Look at the VRF-Demo-DVUplinks

VRF-Demo-DVUplinks-1042

ACTIONS

SummaryMonitorConfigurePermissionsPortsHostsVMs

Port ID

Name

Connectee

Runtime MAC Addr...

Port Group

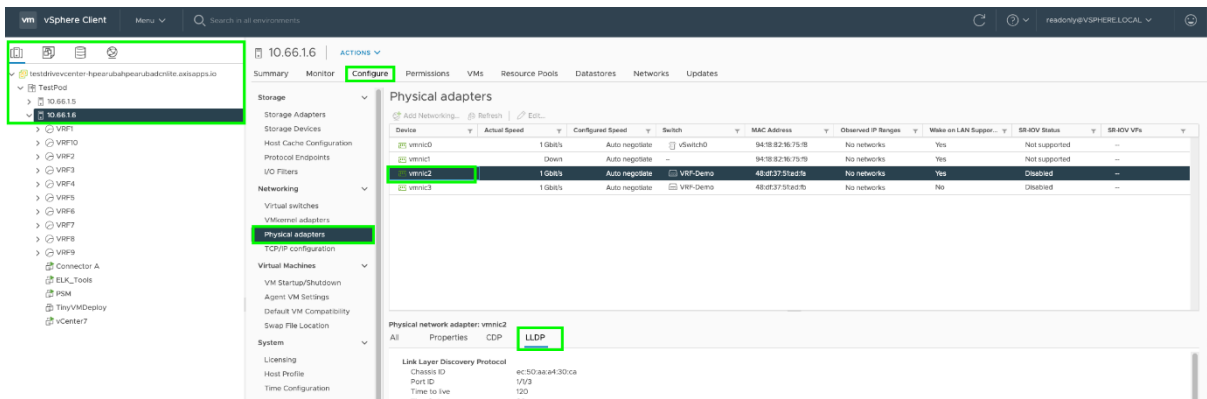
State

VLAN ID

72	dvUplink1	10.66.1.5 - vmnic1	--	VRF-Demo-DVUplin...	Link Up	VLAN trunk: 0-4094
73	dvUplink2	--	--	VRF-Demo-DVUplin...	--	VLAN trunk: 0-4094
74	dvUplink3	--	--	VRF-Demo-DVUplin...	--	VLAN trunk: 0-4094
75	dvUplink4	--	--	VRF-Demo-DVUplin...	--	VLAN trunk: 0-4094
76	dvUplink1	10.66.1.6 - vmnic2	--	VRF-Demo-DVUplin...	Link Up	VLAN trunk: 0-4094
77	dvUplink2	10.66.1.6 - vmnic3	--	VRF-Demo-DVUplin...	Link Up	VLAN trunk: 0-4094
78	dvUplink3	--	--	VRF-Demo-DVUplin...	--	VLAN trunk: 0-4094
79	dvUplink4	--	--	VRF-Demo-DVUplin...	--	VLAN trunk: 0-4094



c. Check the ports allocated and then go to the 10.66.1.6 host and look at the physical host adapters and LLDP config.



What do you see?

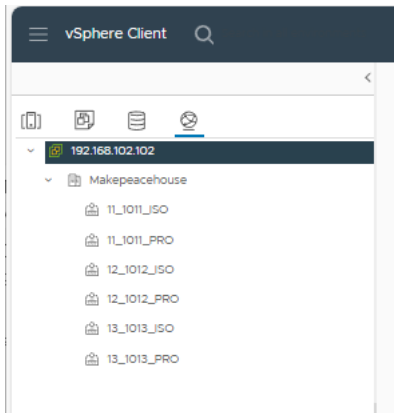
Your vSphere Networking/Physical adapters section should look similar to the above, ask questions if you do not understand or need clarification on anything.

This is the same for other Hypervisors and other physical switches.

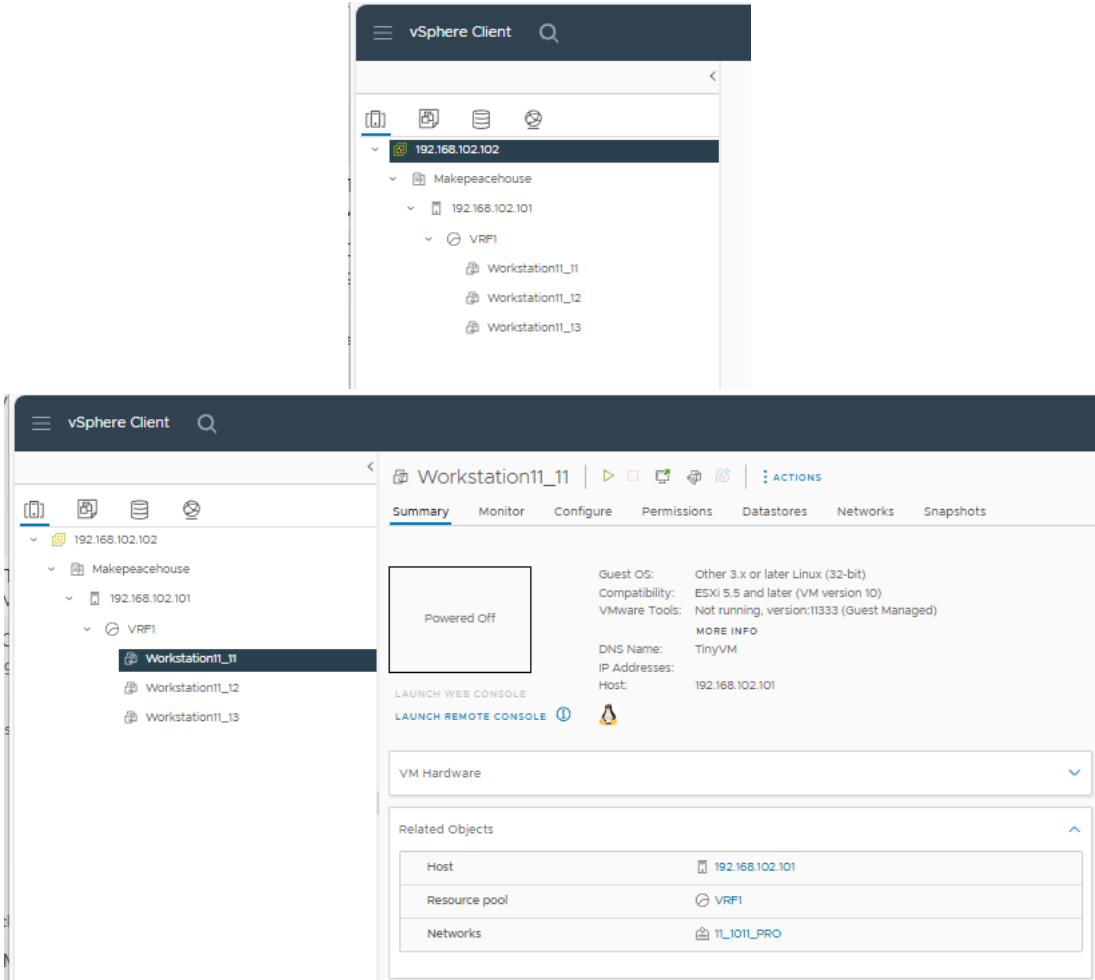
Activity: Access vSphere workloads

The vm console can be accessed via the Vcenter server by selecting your desired vm and launching the Web Console.

1. Login into Vcenter with your vrf<pod>@vsphere. local account. Review the **VMware networking** you have been allocated.



2. Look at the **workloads** you have been allocated and the networks they have been assigned.

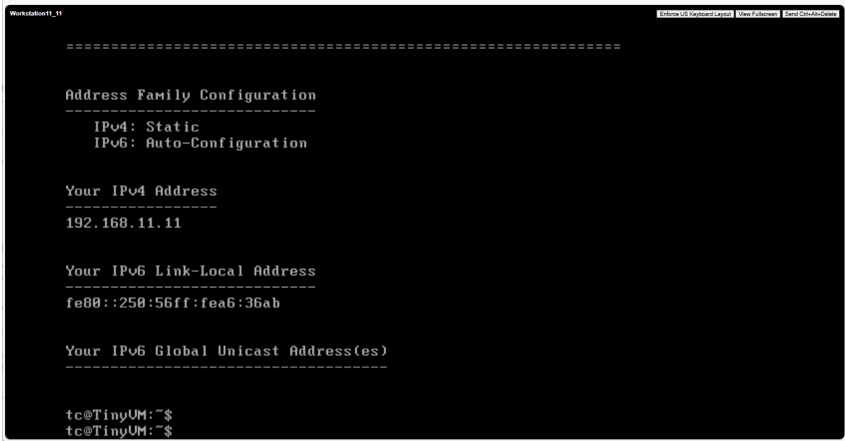


- 3.

In a later step you will be assigning your workloads to different Port-groups, but to start with we recommend leaving them on the default Port Group's, they are bound too.

Activity: Confirm VM Base Networking

If you happened to edit any of the original Port Group configuration, you would also need to edit the IP address of the workloads so they can access the corresponding network. At this point, if no changes have been made to port groups or IP addressing, then the workloads should be able to communicate.



- 1. You can perform simple ping tests to validate this.

Refer to the commands in the section [Commands to change IP](#) and test that you can access the SVI interface on the CX10000 switch and any other workstations.

Testing VM Networking

To validate network connectivity, ensure you can successfully ping both your pods’ VLAN gateway and the IP addresses of the peer VM’s. If connectivity fails, troubleshoot and resolve the network issue before proceeding to the next step.

At this stage, no security policies are enforced, and any traffic restrictions would be most likely due to network or vSphere misconfiguration

Activity: Testing / Exploring in PSM

The URL for the PSM will be provided by the trainer, as it is unique to every lab.

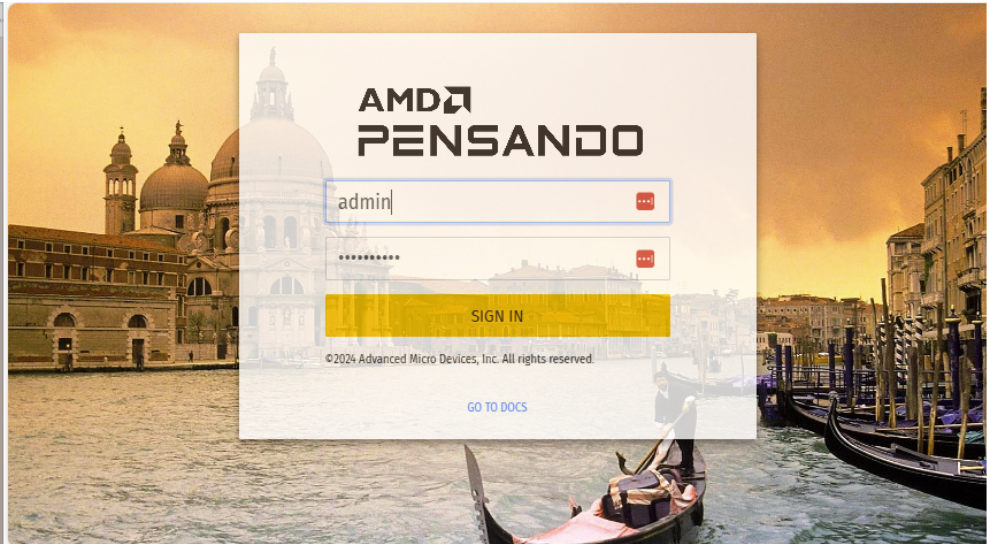
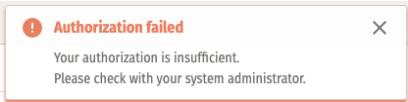
 For more information, options, and a quick walkthrough on PSM, refer to Appendix A or visit the user-guide at <https://networkingsupport.hpe.com/>.

Enable Policy

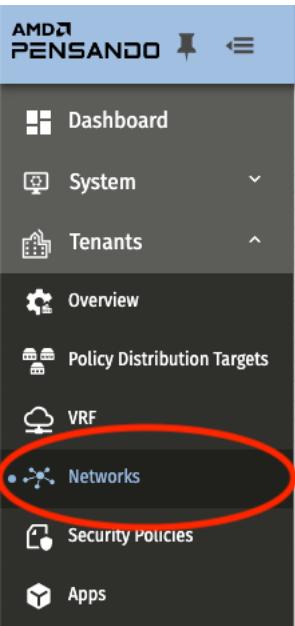
To begin enabling policy enforcement, we need to use the assigned axis pod cross launch link or IP for PSM and place it in a browser to continue to login with your vrf<pod> account.

	Username	Password	Role
	vrf<pod>_user	Pensando0\$	vrf(pod)
Example	vrf1_user	Pensando0\$	Vrf1

⚠ Note: You may see an ‘Authorization failed’ notification in the top-right window corner when working in the PSM GUI. This can be ignored, as it is related to the assigned permissions.



Policy creation and enforcement begins under the Tenant portion of the menu in PSM, specifically under the **Network** section. For the initial phase, we focus on the network section.



Networks

ADD NETWORK

Networks Overview

Networks (54)  

16 Columns  Search Aa Ab R' Q  

☐	VLAN 	Maximum CPS	Maximum Sessions	Allow Session Reuse	Connection Tracking Mode	IP Fragments Forwarding	Service Bypass	Labels	Modification Time
☐	10	Inherited from VRF	Inherited from VRF	Inherited from VRF	Inherited from VRF	enabled	disabled		2025-03-11 09:59:30 GMT+00:00
☐	11	Inherited from VRF	Inherited from VRF	Inherited from VRF	Inherited from VRF	enabled	enabled		2025-03-11 09:59:30 GMT+00:00  
☐	12	Inherited from VRF	Inherited from VRF	Inherited from VRF	Inherited from VRF	enabled	enabled		2025-03-11 09:59:30 GMT+00:00
☐	13	Inherited from VRF	Inherited from VRF	Inherited from VRF	Inherited from VRF	enabled	enabled		2025-03-11 09:59:30 GMT+00:00

1. Select the pencil option on the right end of the line to edit any network configuration.

From the **network** overview section, you can see the policies assigned on ingress or egress, and the service bypass mode is “enabled”.

(All default policies have been created with only an allow rule to get started).

Name

vlan11

Allow Session Reuse

inherit from vrf

IPv4 Ingress Security Policy

Select Security Policy... 

IPv4 Egress Security Policy

vrf1_vlan11_First (default) 

IPv6 Ingress Security Policy

Select Security Policy... 

IPv6 Egress Security Policy

Select Security Policy... 

Ingress Mirror Session

Select Ingress Mirror Session... 

Egress Mirror Session

Select Egress Mirror Session... 

VRF

default

VLAN ID

11

Maximum CPS

Inherit from VRF

Unlimited

Set Value

Maximum Sessions

Inherit from VRF

Unlimited

Set Value

Connection Tracking Mode

inherit from vrf

Service Bypass



IP Fragments Forwarding

inherit from vrf

2. **Disable** “service bypass” for your selected policy and click save.

 **Note:** Enable/Disable service-bypass. When enabled, the distributed services will be disabled. Blue means enabled – meaning no policies will be enforced

Activity: Data Visibility in PSM

Once a policy is enabled, all traffic traversing the switch is subject to policy enforcement. Because logging has already been configured, all corresponding logs will be automatically forwarded to both the PSM and the pre-configured ELK stack for analysis and monitoring.

You can now generate test data between the workloads and out to the internet (if accessible).

Ping works fine to start

```
Workstation11_11
fe80::250:56ff:fea6:36ab

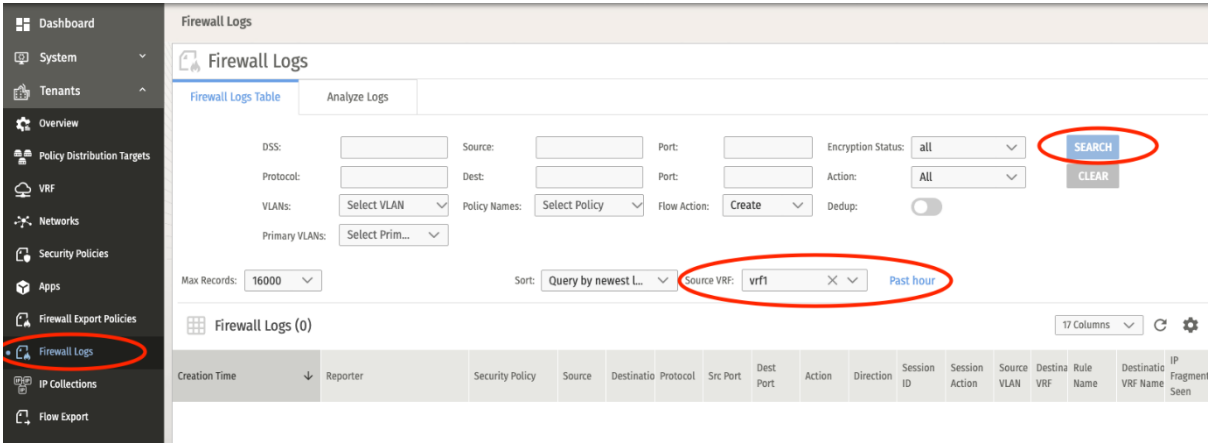
Your IPv6 Global Unicast Address(es)
-----

tc@TinyUM:~$
tc@TinyUM:~$ ping 192.168.11.12
PING 192.168.11.12 (192.168.11.12): 56 data bytes
64 bytes from 192.168.11.12: seq=0 ttl=63 time=11.514 ms
64 bytes from 192.168.11.12: seq=1 ttl=63 time=0.474 ms
^C
--- 192.168.11.12 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.474/5.994/11.514 ms
tc@TinyUM:~$ ping 192.168.11.13
PING 192.168.11.13 (192.168.11.13): 56 data bytes
64 bytes from 192.168.11.13: seq=0 ttl=63 time=12.675 ms
64 bytes from 192.168.11.13: seq=1 ttl=63 time=0.794 ms
^C
--- 192.168.11.13 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.794/6.734/12.675 ms
tc@TinyUM:~$
```

Activity: Viewing logs in PSM

Moving through the left-hand menu in the PSM GUI, go to the **Firewall Logs** section.

- 1. Select Firewall Logs from the menu.
- 2. Select the VRF you are interested in in the drop-down menu.
- 3. Next to the selected VRF, choose the time period you are interested in and apply it to the search.



Do you see any log data?

After disabling the service bypass, the Logs get forwarded to the PSM instance. This happens either every 5 minutes or 10,000 log events, whichever happens first, so there may be a slight delay.

What logs do you see and what do you not see? Why?

- Talk to the group and the trainer to understand what you see and do not see and why.

—

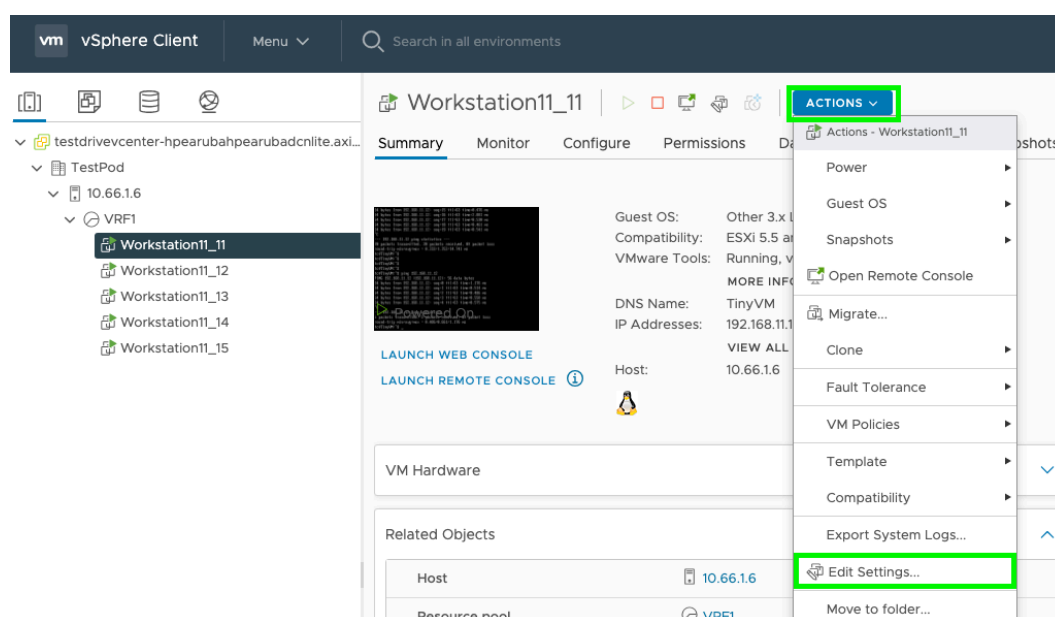
Activity: Configure micro-segmentation

Micro-Segmentation divides the network into smaller segments typically based on individual workloads or applications. This provides you with more granular control over your traffic, while reducing the networks' attack surface and limiting lateral (East-West) movement.

Currently, the attached Port-Group these Workloads have been assigned to is on a promiscuous network, meaning that switching is performed locally within the virtual switch inside the hypervisor and therefore this traffic never needs to leave the ESXi host.

To enable visibility and management of traffic, the workload must be configured to communicate over a PVLAN in isolated mode. This ensures that traffic is forwarded upstream of the host to the switch for inspection.

1. In vSphere Edit the selected **workload** settings.



- 2. Relocate the selected workload vm to your isolated VLAN by browsing the available networks and selecting the isolated version of that network that corresponds to your host’s original network.

Edit Settings
Workstation11_11

Virtual Hardware
VM Options

ADD NEW DEVICE

CPU	1		
Memory	192	MB	
Hard disk 1	80	MB	
Network adapter 1 *	11_1011_PRO 11_1011_ISO Browse ...		☒ Connected
CD/DVD drive 1			☐ Connected
Video card	Specify custom settings		
VMCI device			
Other	Additional Hardware		

Select Network

Filter

Name	NSX Port Group ID	Distributed Switch
11_1011_ISO	--	VRF-Demo
11_1011_PRO	--	VRF-Demo
12_1012_ISO	--	VRF-Demo
12_1012_PRO	--	VRF-Demo
13_1013_ISO	--	VRF-Demo
13_1013_PRO	--	VRF-Demo

6 items

CANCEL

OK

- 3. Repeat this for any or all your workload vm’s and continue to View in PSM

Activity: View Visibility data in ELK

The URL for the ELK stack will be provided by the trainer, as it is unique to every lab.

Logs from the CX10000 can be forwarded to any external syslog collector in real time.

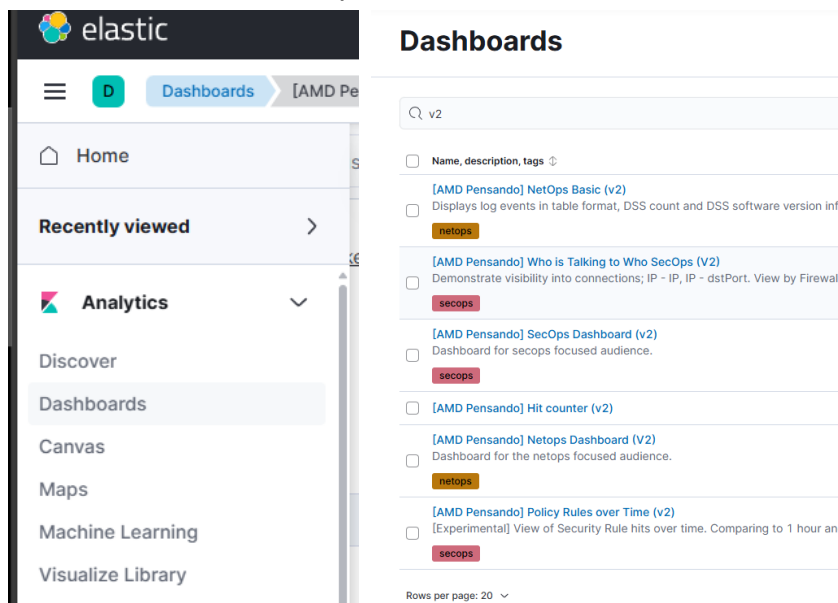
The ELK stack has been selected as the demonstration tool for this lab due to its availability and popularity as a community edition and free open-source suite.

There are many paid and licensed tiers for ELK along with multiple other vendors who provide syslog collectors that can be used in conjunction with the CX10000. You are not tied to using ELK, it is just what we have found to fit best for our lab purposes.

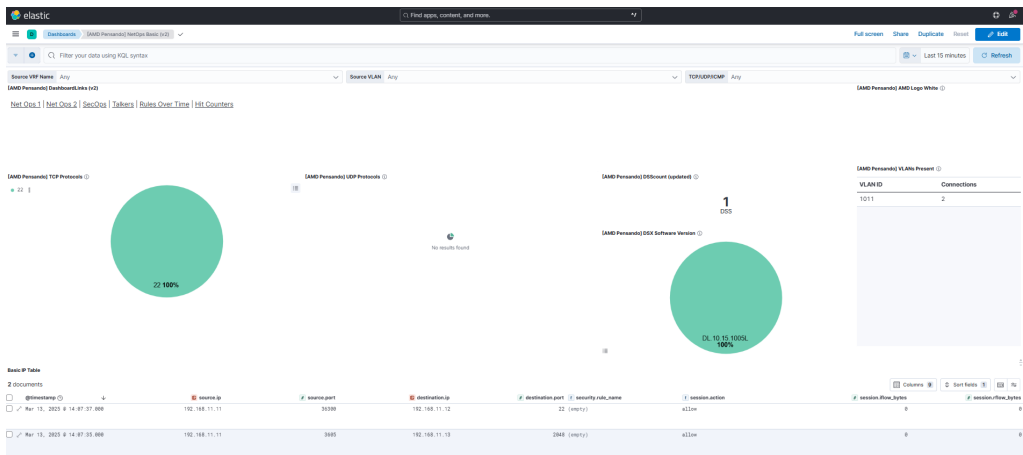
*The ELK stack, like the PSM and Vcenter instances, is shared for all pods.

Username	Password	Role
vrf<pod>_user	Pensando0\$	

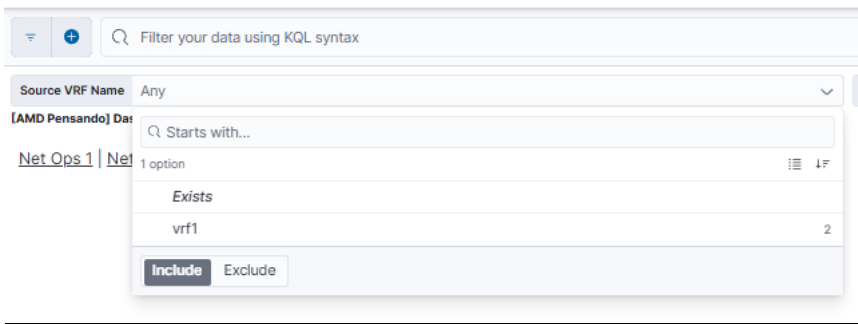
- To access the ELK server, go to the url:
http://<ELK IP>:5601/
(credential: vrf<pod>_user - Pensando0\$)
- Select the **Dashboard** option and search for AMD.



- Select the “[AMD Pensando] NetOps Basic v2” **Dashboard**.



4. Can you see your pod logs? Can you filter the logs to your pods VRF.



- 5. What logs do you see and what do you not see? Why?
- 6. Explore the ELK dashboard by viewing your data through the various dashboards and visualizations. Continue to revisit the dashboards while performing other testing to monitor any changes and gain insights into how your traffic is behaving and what your traffic is doing.

PSM Base Features Testing

There are several base features that exist in PSM regarding firewalling that must be understood to help you effectively apply and construct policies. The tasks we will focus on are:

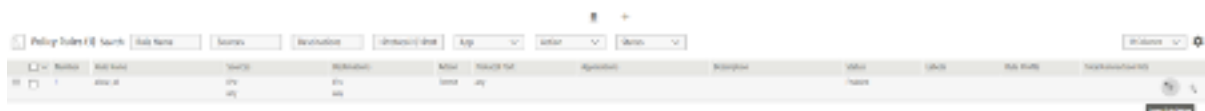
- Block ICMP & Service Bypass
- IP-Fragmentation support
- Connection Tracking
- Session Reuse
- Block SSH
- Block traffic based on IP-Collections
- Block traffic based on VMware tags
- Block IPERF

Task 1 - Block ICMP between two workloads

Make sure you have walked through the previous steps before proceeding.

You will now configure a simple deny ICMP policy between your workload1 and workload2 (directional) in your POD as a test of the next few features.

In the **Security policies** section find your security policy <vrfx_vlanxx_first>, select the pencil on the right side to “Edit Policy”. Under policy Rules click “+” to add a rule to your policy. You may select either the “+” at the top of the rules or the “+” at the end of the line, for the insertion of a rule above or below the existing “allow_all rule”.



Step 1- You need to provide a “Name” for the rule, verify the rule order number is 1, change the action to “Deny”, add a description if needed and type icmp in the protocol box. Enable rule is default

Policy Editing

1 GENERAL 2 SOURCE & DESTINATION 3 PREVIEW [SAVE] [CANCEL]

Rule Name: BLOCK_ICMP_host1_to_2

Description: test blocking ICMP

Rule Order: 1

Action: Deny

Disabled: ☐

Rule Profile: Rule Profile...

Protocol/Ports and Apps

Protocol: icmp

Ports:

Apps: Select Apps...

+ Protocols and Ports

Rule Labels

key value + ADD

There are also additional options around the use of port, protocol or Pre-defined Apps that may be used. (Apps are L4 based on port and protocol only).

Step 2 - In the policy editing window define the source IP and destination IP of your chosen workloads. You have the option to use only the IP or a CIDR range from 0 – 32 if you wish.

Policy Editing

1 GENERAL 2 SOURCE & DESTINATION 3 PREVIEW [SAVE] [CANCEL]

Source and Destination

* Enter at least one option in both source and destination fields.

Source Workload Groups: Select Source Workload Groups...

Destination Workload Groups: Select Destination Workload Groups...

Source IP Addresses: 192.168.11.11

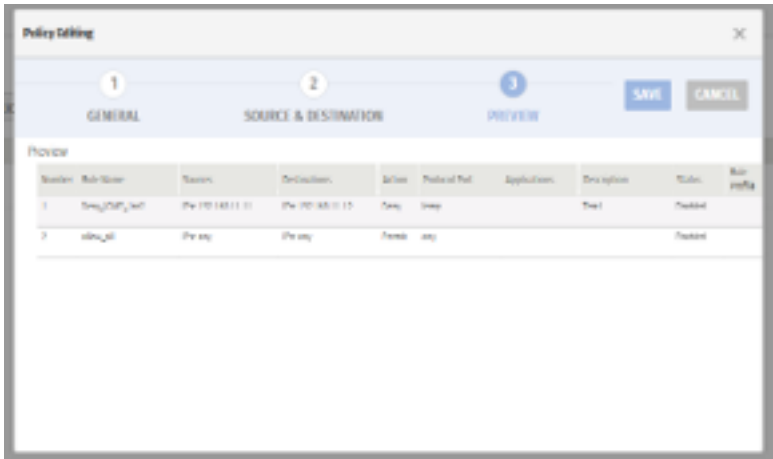
Destination IP Addresses: 192.168.11.12

Source IP Collections: Select Source IP Collections...

Destination IP Collections: Select Destination IP Collections...

Step 3 - Preview the rule to make sure it meets the requirements and click Save and Save policy at the top of the screen.

- Proceed to test if you can ping from host 1-2 and 2-1. Also review the logs in PSM and ELK
- Do they match what you expect?



 Learning Check Task 1

1. Navigate and Overview of the PSM Gui
 2. Explore Security Policies
 3. Created a security policy and all its attributes
 4. Verified a working policy rule
 5. Verified log collection on PSM and ELK

Task 2 - PSM service Bypass

Service bypass is a feature that can be enabled to allow bypass of a VLAN from policy management.

Continuing from the Block-ICMP rule we just created in the previous section. From the **Networks section** go to your vlan and enable the service bypass slider by toggling it to ON (it will turn blue), click save and ok on pop up and then re-run the ping tests and check/view the logs.

Name vlan11	Allow Session Reuse inherit from vrf
IPv4 Ingress Security Policy Select Security Policy... X v	IPv4 Egress Security Policy vrf1_vlan11_First (default) X v
IPv6 Ingress Security Policy Select Security Policy... X v	IPv6 Egress Security Policy Select Security Policy... X v
Ingress Mirror Session Select Ingress Mirror Session... X v	Egress Mirror Session Select Egress Mirror Session... X v
VRF vrf1	VLAN ID 11
Maximum CPS Inherit from VRF Unlimited Set Value	Maximum Sessions Inherit from VRF Unlimited Set Value
Connection Tracking Mode inherit from vrf	Service Bypass ☒ IP Fragments Forwarding inherit from vrf

 **Note:** Enable/Disable service-bypass. When enabled, the distributed services will be disabled.

- Is the output as expected?
 - Are you seeing Logs
 - Is the Enforcement correct



Learning Check Task 2

1. Using the PSM GUI to verify a policy
2. Working with service bypass feature
3. Verifying log visibility

Task 3 - PSM Fragmentation

Any firewall by default will drop fragmented packets, In PSM you can enable support of fragmentation to allow fragmented packets. This is helpful in scenarios such as the early stages of deployment while you are still gaining visibility and learning about what is going on in the network.

To test this, you will send Jumbo ping packets before and after you set this flag to validate the feature.

Step 1 - From your **VM 1** #ping (workload2-ip) -s 9000

- What happens?

Step 2 - From the **Networks** section go to your vlan and enable IP fragment forwarding in the drop-down menu, click save and ok on pop-up.

Name

vlan11

IPv4 Ingress Security Policy

Select Security Policy...

✕

▼

IPv6 Ingress Security Policy

Select Security Policy...

✕

▼

Ingress Mirror Session

Select Ingress Mirror Session...

✕

▼

VRF

vrf1

Allow Session Reuse

inherit from vrf

IPv4 Egress Security Policy

vrf1_vlan11_First (default)

✕

▼

IPv6 Egress Security Policy

Select Security Policy...

✕

▼

Egress Mirror Session

Select Egress Mirror Session...

✕

▼

VLAN ID

11

Maximum CPS

Inherit from VRF

Unlimited

Set Value

Maximum Sessions

Inherit from VRF

Unlimited

Set Value

Connection Tracking Mode

inherit from vrf

Service Bypass

IP Fragments Forwarding

enable

Step 3 - From your VM 1 #ping (workload2-ip) -s 9000

- What happens?

Now check the logs in both PSM and ELK and notice the fragmentation flag is set.

Firewall Logs (1,000)															15 Columns	
Creation Time	Reporter	Security Policy	Source	Destination	Protocol	In Port	Out Port	Action	Direction	Session ID	Source Action	Source VLAN	Destination VRF	Data Source	Destination VRF Name	IP Fragmentation
2024-03-12 12:25:18.000+00:00	Switch Name: vcs10a04-0403 vcs10a04-0403 vcs10a04-0403	Default	10.81.53.124	10.81.53.122	TCP	8071	8081	Allow	From Host	530181	Allow	130	BL18	Any_vrf_vrf	BL18	False
2024-03-12 12:25:18.000+00:00	Switch Name: vcs10a04-0403 vcs10a04-0403 vcs10a04-0403	Default	10.81.53.122	10.81.53.124	TCP	8071	8081	Allow	From Host	530181	Allow	130	BL18	Any_vrf_vrf	BL18	False

Using the PSM GUI to verify a policy

Learning Check Task 3

2. Working with the IP Fragments feature

3. Verify ping and log visibility

Task 4 - PSM Connection Tracking (CT)

DSM modules enforce stateful firewall policies. However, when transitioning from no policy to full enforcement, starting in stateless mode can help. It lets you drop traffic with minimal disruption while monitoring network behavior.

To use stateless mode, set the connection tracking state to 'Disabled' instead of the default 'Enabled'. In this mode, rules act like traditional ACLs, so you may need to define traffic in both directions for bidirectional communication.

Before making any changes to stateless mode:

1. Enable service bypass from your vlan and create a new rule from your security policy to block SSH. (apply on egress on the network). (click Save)
2. Connect a SSH session between workload 1 and workload 2.

User	Password
tc	VMware1!

#ssh tc@192.168.x.x

3. With connection tracking enabled, disable service bypass, what happens to the SSH session. (Save)
4. Re-enable service bypass. (Save)

Change to stateless. (disable connection tracking).

Name vlan11	Allow Session Reuse inherit from vrf
IPv4 Ingress Security Policy Select Security Policy... X v	IPv4 Egress Security Policy vrf1_vlan11_First (default) X v
IPv6 Ingress Security Policy Select Security Policy... X v	IPv6 Egress Security Policy Select Security Policy... X v
Ingress Mirror Session Select Ingress Mirror Session... X v	Egress Mirror Session Select Egress Mirror Session... X v
VRF vrf1	VLAN ID 11
Maximum CPS Inherit from VRF Unlimited Set Value	Maximum Sessions Inherit from VRF Unlimited Set Value
Connection Tracking Mode disable	Service Bypass ☒ IP Fragments Forwarding enable

5. Connect a SSH session between workload 1 and workload 2.
6. With connection tracking disabled, disable service bypass, what happens to the SSH session. (Save)
7. Re-enable service bypass. (Save)



Learning Check Task 4

1. Using the PSM GUI to verify a policy
2. Working with the Service Bypass feature
3. Working with the Connection Tracking feature
4. Testing SSH connectivity

Task 5 - Block IPERF traffic based on IP-Collection

An IP collection is a list of IP addresses that can be referenced at a later time in a firewall policy. This makes policy creation simpler and allows for easier changes to policy in the future. We will add IP Collections for your selected POD Networks.

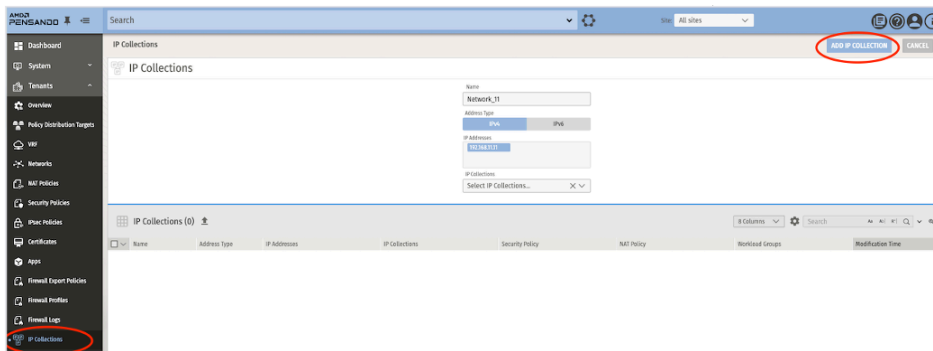
From your PSM:

In this scenario, the IP Collections have **already been preconfigured** on the PSM instance for each specific POD, based on its associated VRF and VLAN. If you were to create them manually, you would follow the steps outlined below.

In the **PSM** GUI move to the **Tenants - IP Collections –Add IP collection** section.

- Provide a name for the Network = **Network_XX**
- Add the correct IP Addresses of your workloads = **192.168.xx.xx**
- Leave “IP Collections” field blank
- Submit by clicking “Add IP Collection” at top of page
- Click “Save Policy” at the top to deploy the changes

Repeat for additional Networks



iPerf is a tool for active measurements of the maximum achievable bandwidth and throughput (UDP/TCP) on IP networks between a sender and a receiver, as well as other abilities.

To test this, we will use 2 hosts **WS1 & WS2** in your respective environment and one will be used as a sender and the other as a receiver.

We will now configure a simple deny IPERF policy between your workload1 (Server) and workload2 (client) using an IP collection to test this specific feature.

Before creating any policy:

- Proceed to test if you can start the iperf server on WS1 and initiate an iperf connection on port 5201 from WS2. (Reference the commands in the “from your Host” section below and run them from your WorkStation)
- Review the logs in PSM and ELK
- Do they match what you expect?

From your PSM:

In the **PSM** GUI move to the **Tenants - Security Policy** section. Hover over the existing policy named <vrfx_vlanxx_first> and select the pencil on the right side to “Edit Policy on Details Page”. Click the “+” to Add a rule to our policy.

Step 1

- Provide a name for this rule “Deny-iperf-vlan x” (vlan x is one of your choosing from your pod)
- Provide an Action of type “Deny”
- Set Rule Order to 1
- Set App to **iPerf** or type the protocol tcp/port 5201 in the appropriate box

The screenshot shows the 'Policy Editing' window with the 'GENERAL' tab selected. The 'Rule Name' field contains 'Deny-iperf-vlan11'. The 'Rule Order' is '303' and the 'Action' is 'Deny'. The 'Description' is 'Deny IPERF'. The 'Disabled' toggle is turned off. The 'Rule Profile' is set to 'Rule Profile...'. In the 'Protocol/Ports and Apps' section, there are two options separated by 'OR': 'tcp' on port '5201' and 'iperf' on port '5201'. The 'Rule Labels' section is empty.

Step 2

- Proceed to section 2 for “Source and Destination”
- In the drop down change the Destination and Source IP Collections to be the statically configured Network_xx ip collection= created above

Policy Editing

1 GENERAL 2 SOURCE & DESTINATION 3 PREVIEW

SAVE CANCEL

Source and Destination

* Enter at least one option in both source and destination fields.

Source Workload Groups

Select Source Workload Groups... X v

Destination Workload Groups

Select Destination Workload Groups... X v

Source IP Addresses

Destination IP Addresses

Source IP Collections

Network_11 X v

Destination IP Collections

Network_11 X v

Step 3

- Preview the new Rule in “vrfx_vlanxx_first”
- Click Save
- Click “Save Policy” at the top to deploy the change.

After policy creation:

- Proceed to test if you can start the iperf server on WS1 and initiate an iperf connection on port 5201 from WS2. (Reference the commands in the “from your Host” section below and run them from your WorkStation)
- Review the logs in PSM and ELK
- Do they match what you expect?

From your hosts:

iperf3 is already installed on the workstations and you can verify this by running the following command from the VM’s console-

```
#iperf3 -v
```

⚠ Iperf3 listens on port 5201 by default

To test

From **Workstation** #1 we will set up iperf server to run and start listening on port 5201 with the following command:

```
#iperf3 -s -p 5201
```

From **Workstation** #2 we will initiate an iperf connection on port 5201 to see if the connection is allowed with the following command:

```
# iperf3 -c 192.168.x.x -t 10 -b 1000000000 -p 5201
```

⚠ If there are any issues with starting iperf on the server machine such as ‘Unable to start listener...’ you can run the following commands and try again:

```
#ps -ef
```

Find the pid number for the iperf process

```
#sudo kill *PID*
```



Learning Check Task 5

1. Using the PSM GUI to verify policy
2. Verifying IP collections
3. Adding a rule to existing policy for iperf
4. Testing host to host connectivity
5. Verifying log visibility

Task 6 - Block SSH traffic based on VMware Tag's

A *workload group* is a logical grouping of endpoint IP addresses or workload objects based on either one or more workload labels or tags, a list of IP collections, or a mix of both workload labels and IP collection objects. A workload-based label selector can be used only on those workloads/VMs that are dynamically imported from vCenter, with the assumption that the workloads are already tagged in vCenter prior to import.

When workload groups are used in a security policy, it enhances the user experience of defining a policy and bringing out the abstraction of IP address constructs, as policy no longer needs to be based on IP addresses.

From your PSM:

In this scenario, the Workload Groups have **already been preconfigured** on the PSM instance for each specific POD, based on its associated labels and values. If you were to create them manually, you would follow the steps outlined below.

In the **PSM** GUI move to the **Workload – Workload Groups – Add Workload Group** section.

- Provide a name for the Workload Group **Workload_x**
- Select the desired Workload Label Key **vcenter.Demo**
- Select the operator as **equals**
- Select the value as **Workloadx**

You can add additional Labels if required.

- Select the desired Workload Label Key **vcenter.VRFs**
- Select the operator as **equals**
- Select the value as **app**
- Click “Save Policy” at the top to deploy the changes.

Dashboard
System
Tenants
Workload
Workload Objects
Workload Groups
Monitoring
Troubleshoot
Orchestrator

Workload Groups

ADD WORKLOAD GROUP

Name
Workloadx

Associated Workload Labels Selector

Workload Label Key	Operator	Value
vcenter.Demo	equals	Workload1

+ ADD

IP Collections
Select IP Collections...

Name
Workload1

Associated Workload Labels Selector

Workload Label Key	Operator	Values
vcenter.Demo	equals	Workload1

+ ADD

IP Collections
Select IP Collections...

Name
Workload1

Associated Workload Labels Selector

Workload Label Key	Operator	Values
vcenter.Demo	equals	Workload1
Workload Label Key	Operator	Label Values
namespace		
orch-name.vcenter		
orchestrator.display-name		
vcenter.Demo		
vcenter.VRFs		

In your vSphere environment the VM's already contain the appropriate Tags and attributes ...

SSH

We will now configure a simple deny SSH policy between your workload1 and workload2 (directional) to test this specific feature. To test this, again we will use 2 hosts WS1 & WS2 in your respective environment.

Before creating any policy:

- Proceed to test if you can ssh from host 1-2 and 2-1. Also review the logs in PSM and ELK
- Do they match what you expect?
- Did you verify the service's bypass settings?

From your PSM:

In the **PSM** GUI, move to the **Tenants - Security Policy** section. Hover over the existing policy named <vrfx_vlanxx_first> and select the pencil on the right side to “Edit Policy on Details Page”. Click the “+” to Add a rule to our policy. Under policy Rules click “+” to add a rule to your policy.



Step 1- You need to provide a “Name” for the rule, verify the rule order is above the default_allow, change the action to “Deny”, add a description if needed and type the protocol tcp/port 22 in the appropriate box. Enable rule is default

Policy Editing

1

2

3

SAVE

CANCEL

GENERAL

SOURCE & DESTINATION

PREVIEW

Rule Name

Block_SSH

Rule Order

1

Action

Deny

Description

Test Block SSH

Disabled

☐

Rule Profile

Rule Profile...

Protocol/Ports and Apps

Protocol

tcp

Ports

22

Apps

SSH

+ Protocols and Ports

Rule Labels

key

value

+

ADD

Step 2 - In the policy editing window from the Source Workload Groups dropdown select the previously created workload group

Policy Editing

1

2

3

SAVE

CANCEL

GENERAL

SOURCE & DESTINATION

PREVIEW

Source and Destination

* Enter at least one option in both source and destination fields.

Source Workload Groups

Workload1

Destination Workload Groups

Workload1

Source IP Addresses

Destination IP Addresses

Source IP Collections

Select Source IP Collections...

Destination IP Collections

Select Destination IP Collections...

Step 3 - Preview the rule to make sure it meets the requirements and click Save and Save policy at the top of the screen.

Policy Editing

1

2

3

SAVE

CANCEL

GENERAL

SOURCE & DESTINATION

PREVIEW

Preview

Number	Rule Name	Sources	Destinations	Action	Protocol Port	Applications	Description	Status	Rule Profile
1	BLOCK_ICMP_host1_to_2	IPs: 192.168.11.11	IPs: 192.168.11.12	Deny	icmp		test blocking ICMP	Enabled	
2	allow_all	IPs: any	IPs: any	Permit	any			Enabled	
3	Block_SSH	IPs: 192.168.11.11	IPs: 192.168.11.12	Deny	tcp/22		test block ssh	Enabled	

- Proceed to test if you can ssh from host 1-2 and 2-1. Also review the logs in PSM and ELK
- Do they match what you expect?

39

- Did you check the service bypass settings?



Learning Check Task 6

1. Using the PSM GUI to verify policy
2. Verifying Workload groups
3. Adding a rule to existing policy for iperf
4. Testing host to host connectivity
5. Verifying log visibility

Task 7 – Configure Macro-Segmentation

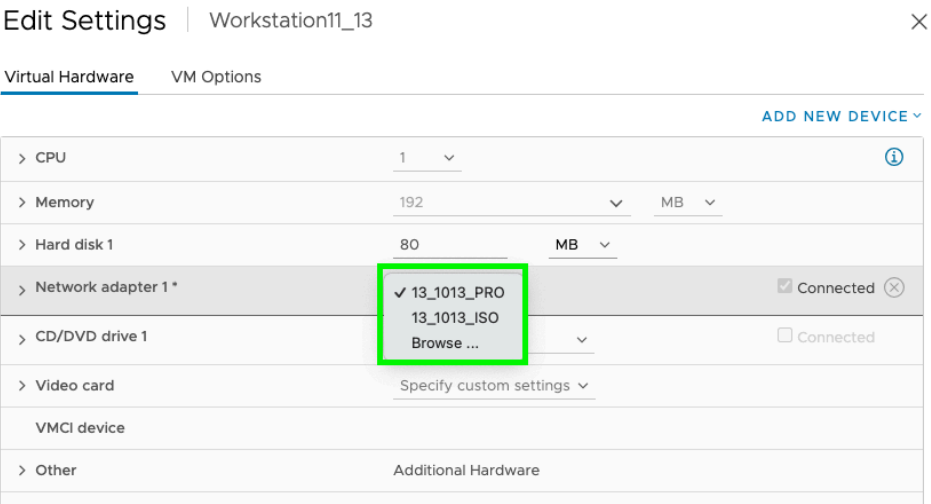
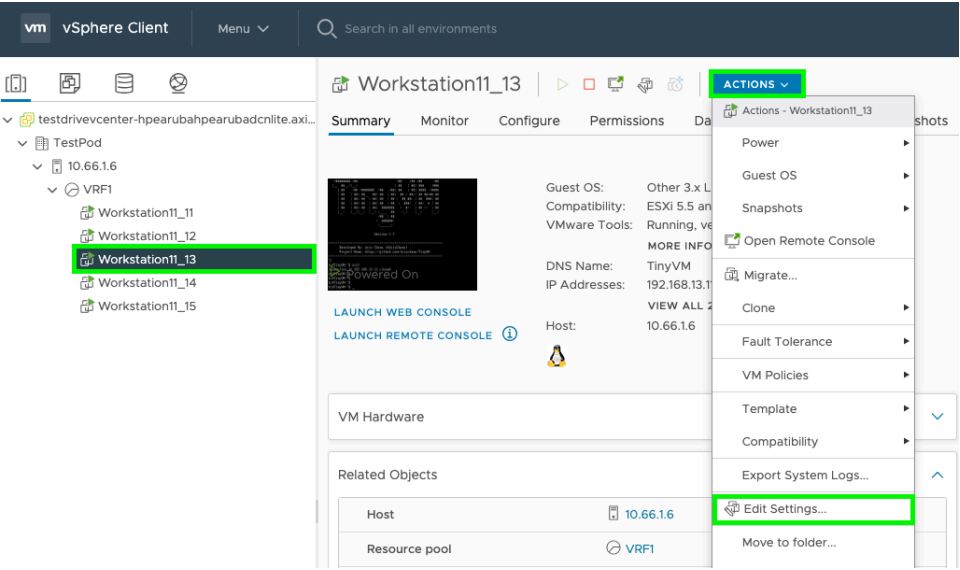
Macro-segmentation is the process of dividing the network into segments typically based on function. This is usually done using VRFs, VLANs or firewall zones. Macro is done between different subnets.

In your vSphere instance, the Workloads were initially assigned to a Port Group on a promiscuous network. This means switching was handled locally within the virtual switch inside the hypervisor, so the traffic never left the ESXi host.

When we walked through the micro-segmentation portion, we relocated the WS1& WS2 to an isolated (<pod>x_10<pod>x_ISO) vlan /network (Port Group in vSphere) for your selected pod to steer traffic up and out to the CX10000.

For this task we have selected WS 4 & 5 in your assigned pod. These workstations have been deployed on a separate network from WS 1-3 with static ip's and configured with the appropriate port group for the chosen environment.

You may Pick a different Isolated port group from the ones that were used in the previous micro segmentation task labeled (<pod>x_10<pod>_PRO, ex. 14_1014_PRO) for your selected pod if needed.



Select Network



Filter

Name	NSX Port Group ID	Distributed Switch
11_1011_ISO	--	VRF-Demo
11_1011_PRO	--	VRF-Demo
12_1012_ISO	--	VRF-Demo
12_1012_PRO	--	VRF-Demo
13_1013_ISO	--	VRF-Demo
13_1013_PRO	--	VRF-Demo

6 items

CANCEL OK

Repeat this for any or all your workload vm's and continue to View in PSM

If needed refer to the section above for the commands required to edit a VM's Ip:
[Commands to edit a VM's IP](#)

You may create any type of new rule or policy here, whether it is blocking or allowing ICMP, SSH, Iperf or anything else you would like to test from the **Security policies** section.

Find your existing security policy <vrfx_vlanxx_first>, select the pencil on the right side to "Edit Policy". Under policy Rules click "+" to add a rule to your policy. You may select either the "+" at the top of the rules or the "+" at the end of the line, for the insertion of a rule above or below the existing "allow_all rule".

Make sure to preview the rules to make sure it meets the requirements and click Save and Save policy at the top of screen.

Proceed to test if you can Ping, SHH, run IPERF or whichever you selected from your selected hosts.

- review the logs in PSM and ELK
- Do they match what you expect?

Depending on how they decide to send traffic ingress /egress may need to add editing policy or vlan



Learning Check Task 7

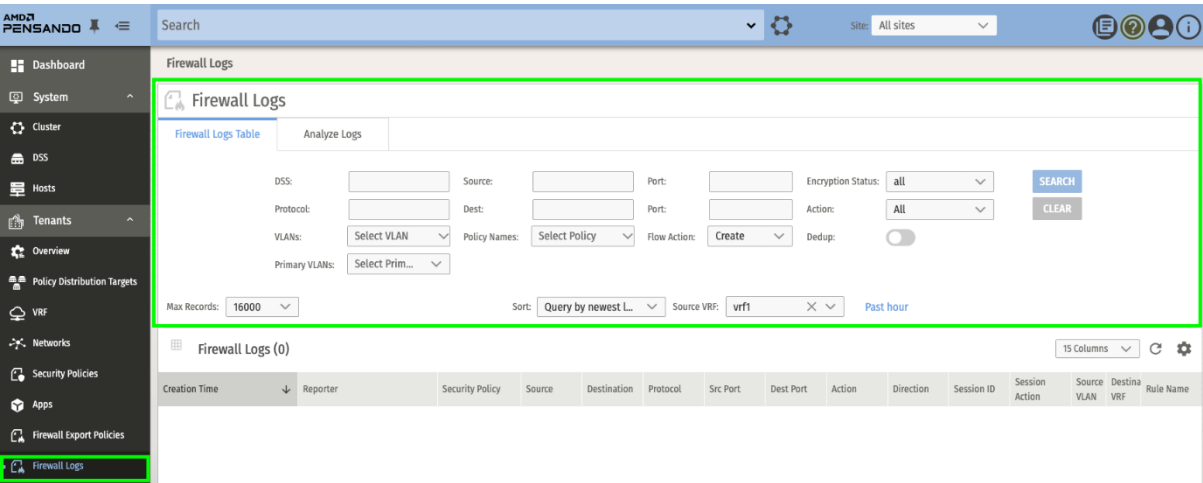
1. Using the PSM GUI to verify policy
2. Editing WS port groups
3. Changing WS IP if required
4. Creating or adding a rule to existing policy
5. Testing host to host connectivity
6. Verifying log visibility

Task 8 – Verifying Logs inside PSM

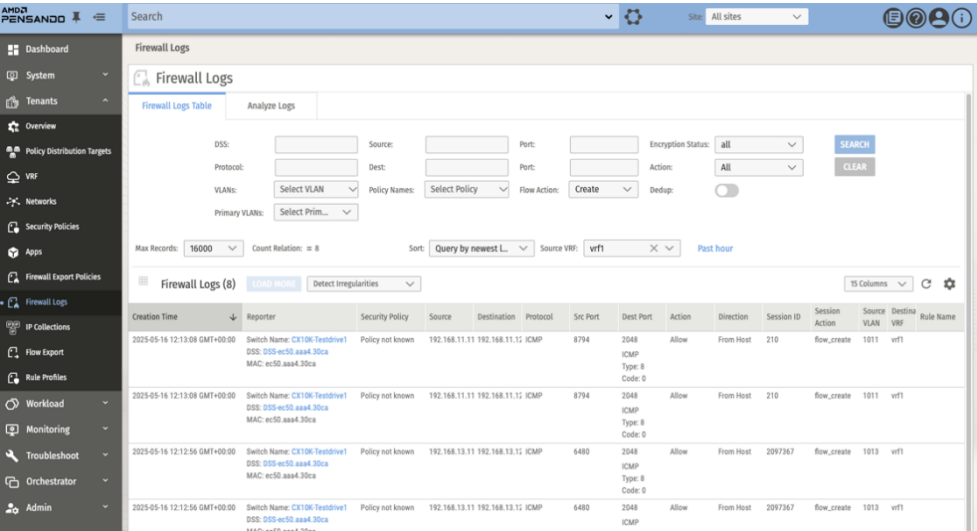
Firewall logs from the DSS can be exported to an external destination/collector, or to the PSM itself. Firewall logs display security policies and event details. They show allowed, denied, and rejected policies along with other important information about the network flows.

You can filter which logs are shown in the Firewall Logs Table in many ways including but not limited to:

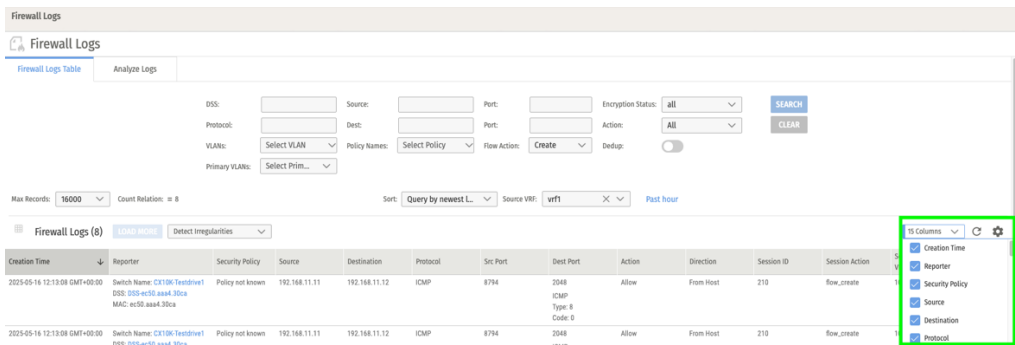
IP, Port, Protocol, Vlan, policy name, VRF and others as seen below



Pick a log and expand it to see the details ...



You can also customize the selected columns to match your specific needs.



⚠ There are scalability limitations when firewall logging is enabled in terms of log export rate, log ingest rate and data retention in the PSM if it is configured as the collector

Learning Check Task 8

1. Using the PSM GUI to verify Logs
2. Editing columns in log output

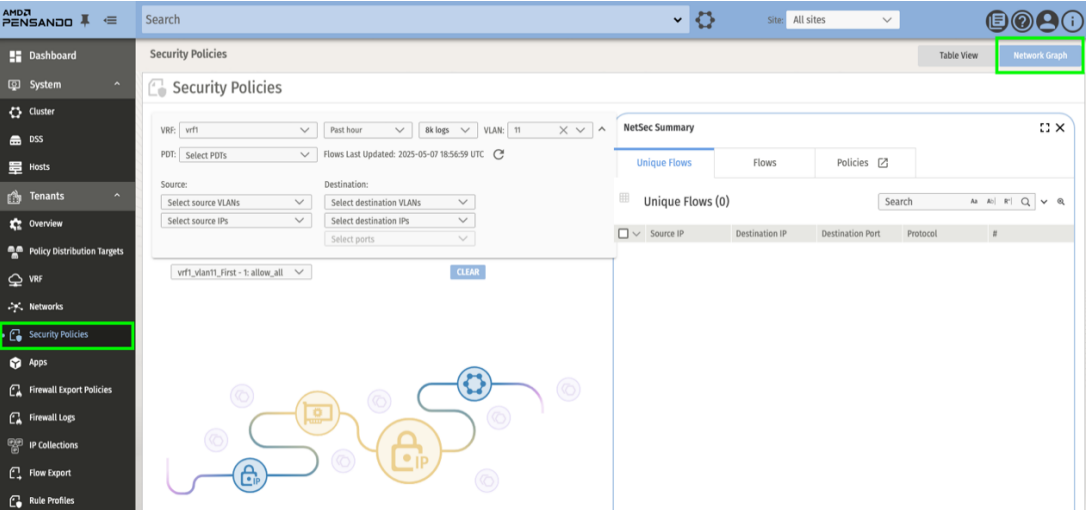
Task 9 – PSM Network Graph Feature

Inside of PSM, you can add security polices in two ways:

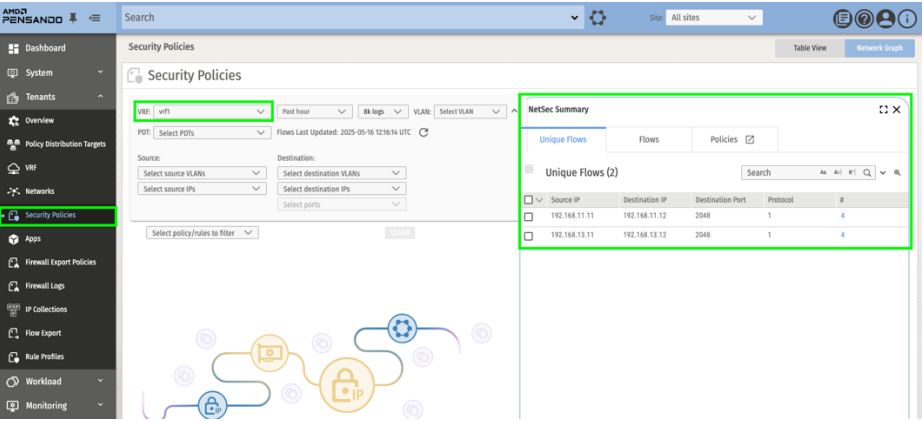
- Manually through the table view
- With the auto discover option using the **network graph** feature

The **Network Graph** method helps discover the relationships between endpoints within a VRF, within and between VLANs, and within or between endpoint groups. Through an iterative process of analyzing the behavior and connections of your endpoints and applications over a dedicated time domain, the network graph method will help discover and automate a zero-trust network policy.

- This feature is only available when PSM is the export target for F/W logs
- Under **Tenants**, locate the **Security Policies** menu item and click on it. From there, proceed to find the **Network Graph** in the upper right-hand corner and select it.



You can now Filter for your pod VRF. Flows should already exist or begin to populate under ‘**Unique Flows**’ in the **NetSec** summary section.



While reviewing the list of unique flows for your assigned POD (VRFx), you can double-click any flow number marked under the # symbol to see its details.

NetSec Summary

Unique FlowsFlowsPolicies

Unique Flows (2)

	Source IP	Destination IP	Destination Port	Protocol	#
☐	192.168.11.11	192.168.11.12	2048	1	4
☐	192.168.13.11	192.168.13.12	2048	1	4

In the ‘**Show flow details**’ pop-up, you can explore key information specific to your flow, including Source, Destination, Port, Protocol, Session ID and more.

Flow Details

Unique Flow Details (4)

	Creation Time	Reporter	Source	Src Port	Destination	Dest Port	Protocol	Action	Session ID	Session Action	Packets Sent	Packets Received	Bytes Sent	Bytes Received	Policy	Rule Name
☐	2025-05-16 12:13	DSS: ec50.aaa4.3	192.168.11.11	8794	192.168.11.12	2048	1	allow	210	flow_create						
☐	2025-05-16 12:13	DSS: ec50.aaa4.3	192.168.11.11	8794	192.168.11.12	2048	1	allow	210	flow_create						
☐	2025-05-16 12:10	DSS: ec50.aaa4.3	192.168.11.11	8788	192.168.11.12	2048	1	allow	2097368	flow_create						
☐	2025-05-16 12:10	DSS: ec50.aaa4.3	192.168.11.11	8788	192.168.11.12	2048	1	allow	2097368	flow_create						

You now create a policy from these unique flows. Simply select the flow you are interested in and click on the ‘+’ icon.

NetSec Summary

Unique FlowsFlowsPolicies

Unique Flows (2) +

	Source IP	Destination IP	Destination Port	Protocol	#
☒	192.168.11.11	192.168.11.12	2048	1	4
☐	192.168.13.11	192.168.13.12	2048	1	4

From here, you can walk through the same steps as in previous tasks for **policy creation**: Enter a “Name” for the rule, verify the rule order, change the action, add a description and more.

You can create any type of rule or policy here, whether it is blocking or allowing ICMP, SSH, Iperf or anything else you would like to test.

Make sure to preview the rules to make sure it meets the requirements and click Save and Save policy at the top of the screen.

- Proceed to test if you can ping, SSH etc. from host 1-2 and 2-1. Also review the logs in PSM and ELK
- Do they match what you expect?



Learning Check Task 9

1. Using the PSM GUI to verify policy
2. Understanding NetSec summary
3. Reviewing specific flow details
4. Creating policy from a flow

Task 10 – PSM Security Policy Evaluation

The Security Policy Evaluation screen is used to view and monitor traffic flow in the security policies, between two IPs - source and destination. Just enter a source and destination IP to generate the flow details between them.

These results can be further filtered based on the source network and destination network (using network name or VLAN ID). However, in order to do this, a specific VRF must be selected first.

⚠ Both the source and destination network will belong to the same VRF

In the **PSM** GUI move to the **Troubleshoot – Security Policy Evaluation** section

1. Provide a Source IP address & Destination IP address
2. Select the desired VRF for your pod
3. Select the Network name VRFxx for your pod
4. Click submit to display results

1. Enter Ip

Security Policy Evaluation

Source IP: 192.168.41.11 Destination IP: 192.168.41.12

Workload: Workload Name Workload: Workload Name

VRF: VRF VRF: VRF

Network: ☐ Name ☐ VLAN-ID ☐ Network Name Network: ☐ Name ☐ VLAN-ID ☐ Network Name

Egress Policy PDFs: Associated PDFs

Ingress Policy PDFs: Associated PDFs

Protocol/Port: Protocol Port

SUBMIT

2. Select VRF

Security Policy Evaluation

Source IP: 192.168.41.11 Destination IP: 192.168.41.12

Workload: Workload Name Workload: Workload Name

VRF: vrf4 VRF: VRF

Network: ☐ Name ☐ VLAN-ID ☐ Network Name Network: ☐ Name ☐ VLAN-ID ☐ Network Name

Egress Policy PDFs: Associated PDFs

Ingress Policy PDFs: Associated PDFs

Protocol/Port: Protocol Port

SUBMIT

3. Select Network

AMNIO
PENSANDO

Search

Site: All sites

Dashboard

System

Tenants

Workload

Monitoring

Alerts & Events

Troubleshoot

Security Policy Evaluation

Security Policy Evaluation

Source IP: 192.168.41.11

Destination IP: 192.168.41.12

Workload: Workload Name

Workload: Workload Name

VRF: vrf4

VRF: vrf4

Network: ☒ Name ☐ VLAN-ID vlan41

Network: ☒ Name ☐ VLAN-ID vlan41

Egress Policy PDTs:

Associated PDTs

Ingress Policy PDTs:

Associated PDTs

Protocol/Port:

Protocol

Port

SUBMIT

4. Expected Output

AMNIO
PENSANDO

Search

Site: All sites

Dashboard

System

Tenants

Workload

Monitoring

Alerts & Events

Troubleshoot

Security Policy Evaluation

NAT Policy Evaluation

Mirror Sessions

Orchestrator

Center

Admin

Security Policy Evaluation

Source IP: 192.168.41.11

Destination IP: 192.168.41.12

Workload: Workload Name

Workload: Workload Name

VRF: vrf4

VRF: vrf4

Network: ☒ Name ☐ VLAN-ID vlan41

Network: ☒ Name ☐ VLAN-ID vlan41

Egress Policy PDTs:

Associated PDTs

Ingress Policy PDTs:

Associated PDTs

Protocol/Port:

Protocol

Port

SUBMIT

EXPORT JSON

Security Policies

Firewall Logs

192.168.41.11

Source Network
Name: vlan41
Egress Policy: vrf_vlan41_First

Source VRF
Name: vrf4
Egress Policy: No Policy Associated

Destination VRF
Name: vrf4
Ingress Policy: No Policy Associated

Destination Network
Name: vlan41
Ingress Policy: No Policy Associated

192.168.41.12

Security Policy Results (1)

Source Network	Egress Network Policy	Source VRF	Egress VRF Policy	Destination VRF	Ingress VRF Policy	Destination Network	Ingress Network Policy	Action
vlan41	vrf4_vlan41_First Propagation Complete. Updated on 2 PDT: default Status: Propagation Complete Updated: 2	vrf4	No Policy Associated	vrf4	No Policy Associated	vlan41	No Policy Associated	Permit

Additionally, the results can be filtered based on protocol/port (entered in the same format) and Workload (workload name can be entered or selected from the dropdown options). Expanding each row shows the Egress/Ingress Network policy and Egress/Ingress VRF policy details.

Security Policies

Firewall Logs

192.168.41.11

Source Network
Name: vlan41
Egress Policy: vrf_vlan41_First

Source VRF
Name: vrf4
Egress Policy: No Policy Associated

Destination VRF
Name: vrf4
Ingress Policy: No Policy Associated

Destination Network
Name: vlan41
Ingress Policy: No Policy Associated

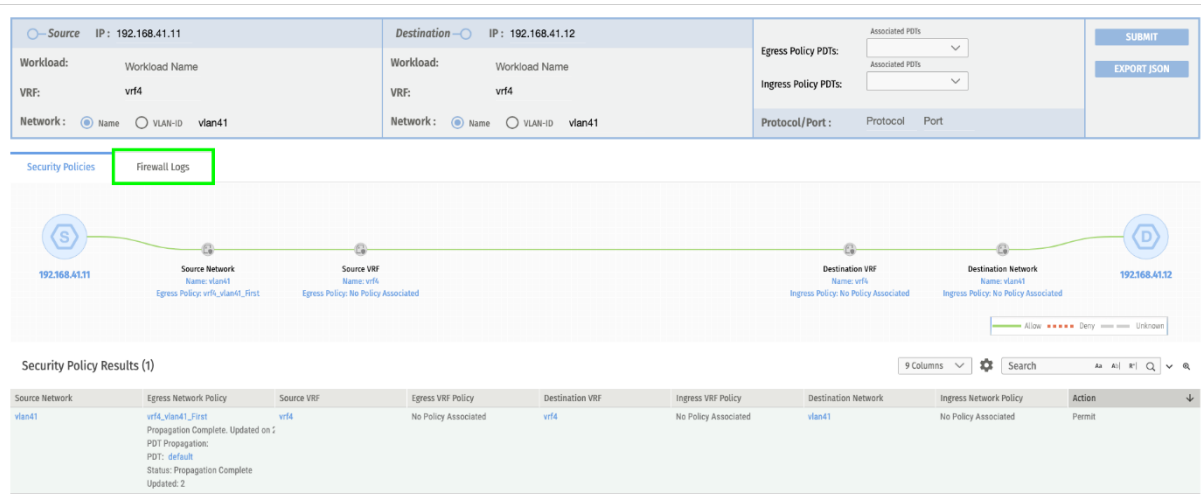
192.168.41.12

Security Policy Results (1)

Source Network	Egress Network Policy	Source VRF	Egress VRF Policy	Destination VRF	Ingress VRF Policy	Destination Network	Ingress Network Policy	Action
vlan41	vrf4_vlan41_First Propagation Complete. Updated on 2 PDT: default Status: Propagation Complete Updated: 2	vrf4	No Policy Associated	vrf4	No Policy Associated	vlan41	No Policy Associated	Permit

Show Details

The firewall logs tab shows the firewall logs corresponding to the inputted IPs/criteria and the evaluation details can be exported to a .json file by using the “Export JSON” button.



Learning Check Task 10

1. Using the PSM GUI to verify policy Evaluation
2. Viewing traffic flows between WS
3. Understanding Allowed and denied flow visualization

Future Tasks:

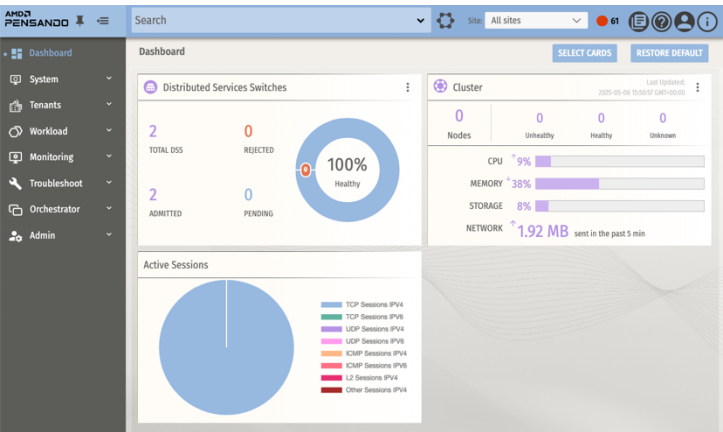
Summary

This guide offers a high-level overview of CX 10000’s capabilities. It was designed to help you explore and experiment with vSphere, PSM, and ELK in a controlled lab environment. The goal was to get you familiar with what you can configure and manage through these tools. Hopefully this provides you with the opportunity to test, learn, and deepen your understanding of the entire solution. Feel free to reach out with any issues, questions, additional use-cases or recommended improvements and keep an eye out for the survey that will be sent out after the lab is completed.

Appendix A: Exploring the PSM GUI

The PSM dashboard provides an extensive list of all the major components inside the PSM platform. This lab includes many pre-configured options to help you quickly access the core features we are trying to showcase for PSM. Although there are options that are not covered in this section for those that are curious, these are items such as VRF creation, Flow export Policy, Firewall Export Policy, Policy Distribution Targets, IP Collections, APP creation and Workload Groups. More detailed configuration information on all these items can be found in the online user guide linked below.

After logging in to the PSM the Main dashboard appears. This view is fully customizable, allowing administrators to choose what they would like to view upon initial login. The left-hand menu lists multiple options to select from based on the task required.



Starting in the **System** section of the menu:

The **cluster** view provides details of the health of the nodes and additional utilization details for each of the nodes in the cluster. In the **DSS** section, you can verify the total

number of Switches and DSM's (DPU's) admitted to the cluster as well as health, version, IP etc.

The **Hosts** section shows you which physical ESXi servers (in the case of using VMware) are hosting the PSM instance.

For our lab today there should be a total of 2 DSS and 4 DSM's showing healthy and admitted. If the DSS are not healthy, contact the lab administrator. You cannot proceed with the lab unless they appear as in the following diagram.

Dashboard

System

Cluster

DSS

Hosts

Tenants

Workload

Monitoring

Distributed Services Switches

Distributed Services Switches Overview

Distributed Services Switches (2)

16 Columns

Search

AA Aa R T Q

☐	Switch Name	DSS-ID	DSM	Switch OS Version	Switch Serial #	Management	Health	Labels	Flow Export Policies	FW Logging Export	Policy Scale	Forwarding Profile	Associated Policy Distribution Targets	Site
☐	CX10K-Testdriv	DSS-ec50.aaa4.30ca	DSM 1/1: ec50.aaa4.312a DSM 1/2: ec50.aaa4.313a	DL 10.15.1010	US2AL6801F	Mode: Out of Band IP address: 10.66.1.31			Elastiflow	ELK_AND_PSM	Max 6K IPv...	Leaf	default	
☐	CX10K-Testdriv	DSS-ec50.aaa4.30b3	DSM 1/1: ec50.aaa4.3113 DSM 1/2: ec50.aaa4.3123	DL 10.15.1010	US2AL68030	Mode: Out of Band IP address: 10.66.1.32			Elastiflow	ELK_AND_PSM	Max 6K IPv...	Leaf	default	

You will spend most of your time in the Tenant section of the PSM menu. It contains 15 sub-sections. For this lab, we will only be covering the portions you will use during the testing and exploring of the defined tasks we have set up. To learn more, use the PSM Help in the GUI or check the 'Graphical User Interface' section of the online user guide located here ->

https://arubanetworking.hpe.com/techdocs/Pensando/AMD_Pensando_PSM_for_DSS_Guide-1.100.2-T.pdf

Tenants

Overview

Policy Distribution Targets

VRF

Networks

Security Policies

Apps

Firewall Export Policies

Firewall Logs

IP Collections

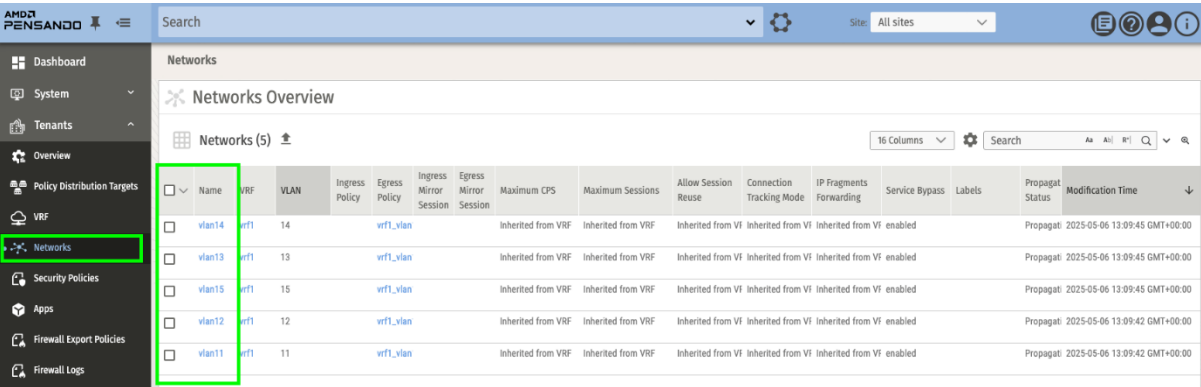
Flow Export

Rule Profiles

Under **Tenants**, locate the **Networks** section and click on it. The Network section provides two main functions

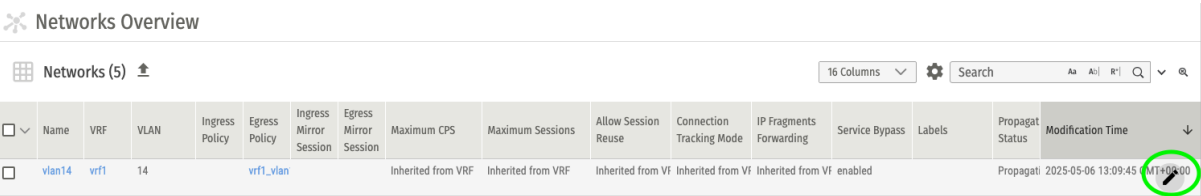
- The ability to add new networks
- The ability to edit and update the various settings for existing networks

In the **Networks** Overview there should be five networks with a naming convention of vlanxx depending on your assigned vrf_user id which you will be assigned.



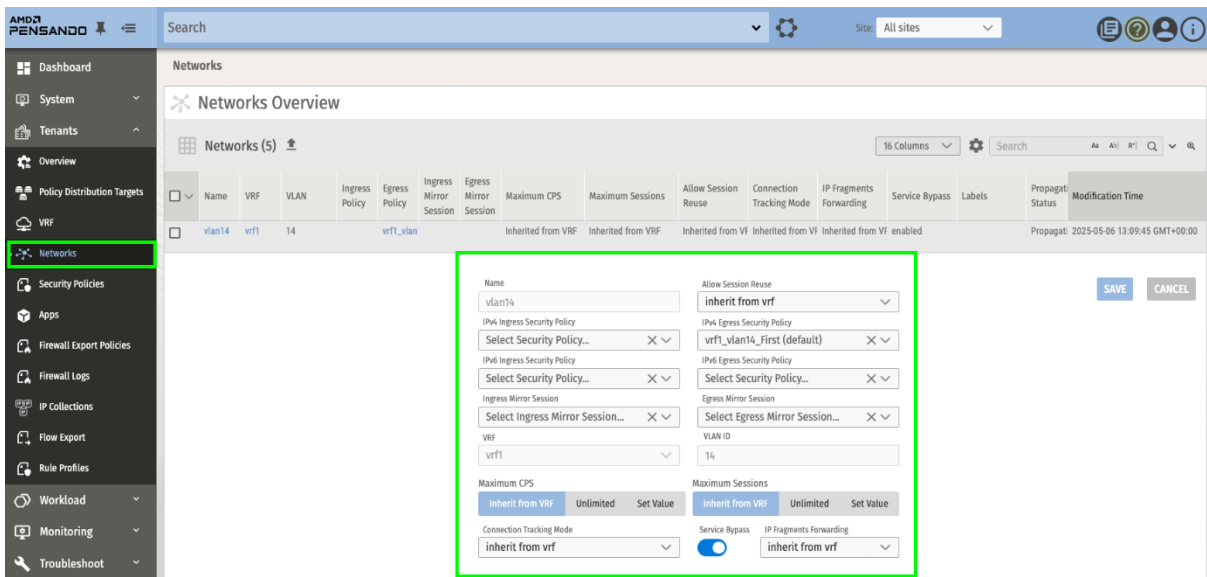
Networks															
Networks Overview															
Networks (5)															
	Name	VRF	VLAN	Ingress Policy	Egress Policy	Ingress Mirror Session	Egress Mirror Session	Maximum CPS	Maximum Sessions	Allow Session Reuse	Connection Tracking Mode	IP Fragments Forwarding	Service Bypass	Labels	Modification Time
☐	vlan14	vrf1	14		vrf1_vlan			Inherited from VRF	Inherited from VRF	Inherited from VI	Inherited from VI	Inherited from VI	enabled		Propagati: 2025-05-06 13:09:45 GMT+00:00
☐	vlan13	vrf1	13		vrf1_vlan			Inherited from VRF	Inherited from VRF	Inherited from VI	Inherited from VI	Inherited from VI	enabled		Propagati: 2025-05-06 13:09:45 GMT+00:00
☐	vlan15	vrf1	15		vrf1_vlan			Inherited from VRF	Inherited from VRF	Inherited from VI	Inherited from VI	Inherited from VI	enabled		Propagati: 2025-05-06 13:09:45 GMT+00:00
☐	vlan12	vrf1	12		vrf1_vlan			Inherited from VRF	Inherited from VRF	Inherited from VI	Inherited from VI	Inherited from VI	enabled		Propagati: 2025-05-06 13:09:42 GMT+00:00
☐	vlan11	vrf1	11		vrf1_vlan			Inherited from VRF	Inherited from VRF	Inherited from VI	Inherited from VI	Inherited from VI	enabled		Propagati: 2025-05-06 13:09:42 GMT+00:00

To edit any of the required columns, simply hover over the desired network and click on the pencil icon.



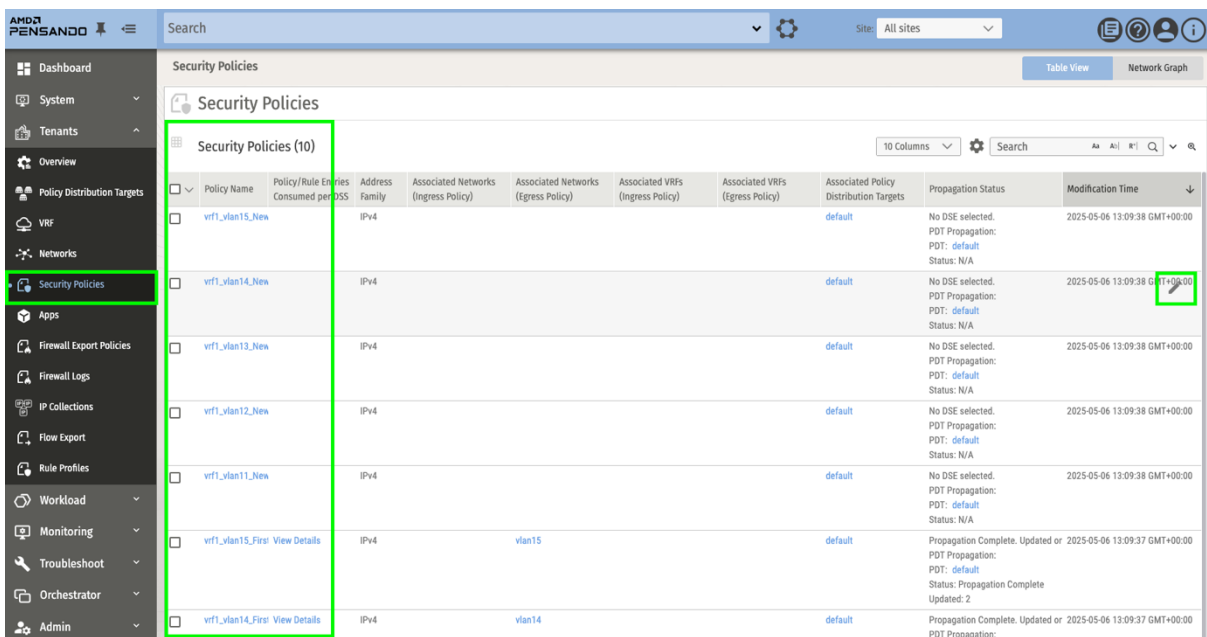
Networks Overview															
Networks (5)															
	Name	VRF	VLAN	Ingress Policy	Egress Policy	Ingress Mirror Session	Egress Mirror Session	Maximum CPS	Maximum Sessions	Allow Session Reuse	Connection Tracking Mode	IP Fragments Forwarding	Service Bypass	Labels	Modification Time
☐	vlan14	vrf1	14		vrf1_vlan			Inherited from VRF	Inherited from VRF	Inherited from VI	Inherited from VI	Inherited from VI	enabled		Propagati: 2025-05-06 13:09:45 GMT+00:00

Once inside your selected network, there are various options that can be selected and changed, as shown in the diagram below. You will walk through many of these options in the hands-on tasks later in this guide.



Under **Tenants**, locate the **Security Policies** menu item and click on it. This section is where your network security rules can be created or modified.

In this section you will see multiple pre-configured policies which correspond to your selected POD with a naming convention like vrfx_vlanxx_New & vrfx_vlanxx_First.



To edit any of the required columns, simply hover over the desired policy name and click on the **pencil icon**.

Again, once inside your selected policy there are various options that can be selected and changed as shown in the diagram below.

Policy Editing

1

2

3

GENERAL

SOURCE & DESTINATION

PREVIEW

SAVE

CANCEL

Rule Name

Rule Order

Action

2

Permit

Description

Disabled

Rule Profile

Rule Profile...

Protocol/Ports and Apps

Protocol

Ports

Apps

Select Apps...

+ Protocols and Ports

Rule Labels

key

value

ADD

Policy Editing

1

2

3

GENERAL

SOURCE & DESTINATION

PREVIEW

SAVE

CANCEL

Source and Destination

Enter at least one option in both source and destination fields.

Source Workload Groups

Destination Workload Groups

Select Source Workload Groups...

Select Destination Workload Groups...

Source IP Addresses

Destination IP Addresses

Source IP Collections

Destination IP Collections

Select Source IP Collections...

Select Destination IP Collections...

You will walk through many of these options in the hands-on tasks section in this guide.